

claude-windows / 1a739b96-ed7c-4c3e-8dfa-a085b1fcfab8

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\1a739b96-ed7c-4c3e-8dfa-a085b1fcfab8.jsonl`  
- SHA-256: `c05254d775182938d079e1792d7dd372704f766515d527d7c31525c4a4d6389f`  
- Source modified: `2026-06-22T09:14:06+00:00`  
- Imported at: `2026-07-05T16:48:17+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `1a739b96-ed7c-4c3e-8dfa-a085b1fcfab8`
```

Transcript

1. user (2026-06-18T07:43:02.149Z)

Stand up a NEW tracked subproject (owner-requested, exploratory ? for future sessions to build on): package the local badminton-highlight pipeline as a CROSS-PLATFORM DESKTOP APP (Windows/macOS/Linux), with the prerequisite cross-platform port.

VISION: target users are recreational sports enthusiasts (NOT tech-savvy). Their match videos are huge (GBs of 4K). Asking them to upload to a cloud UI is slow, bandwidth-heavy, privacy-fraught, and bloats their drives. The product surface that fits the "local, cheap, efficient" thesis: a DESKTOP APP where the user PLUGS IN the camera's USB/SD card, clicks once, and gets back ONLY the highlight reels ? fully on-device, no upload, no original-video bloat.

YOUR FIRST DELIVERABLE = create the tracked project doc `docs/DESKTOP\_APP\_PLAN.md` following `docs/PROJECT\_DOC\_TEMPLATE.md` (status header · decisions-locked · §7 progress tracker as source-of-truth · definition-of-done), index it one line in `docs/README.md`, and open a PR. Mark everything `[design]` ? nothing is built; this is a scoping doc. Branch from origin/master per the branching-safety convention in CLAUDE.md / ~/.claude/CLAUDE.md (fetch; checkout -b off origin/master; explicit-path staging; ONE PR).

READ FIRST (you have no memory of the originating conversation): `memory/current-state.md` (live state ? the local pipeline is mature: WASB shuttle detector + improved windowing, runs fully-local via `tools/generate\_highlights.py --skip-ai-handoff`), `docs/PLATFORM\_ARCHITECTURE.md` (the PLANNED StorageBackend/ComputeBackend registries ? the right seam for a "LocalDesktop" compute backend), `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` + `docs/COMMERCIALIZATION.md` (strategy/licensing ? commercial-clean MIT/Apache/BSD only; Gemini = inference/optional only, never required, never a training source), `docs/CODE\_MAP.md`.

THE HARD PART ? make this the doc's centerpiece (it's the real "porting to Linux and Mac" subproject): the detector today runs WASB/TrackNet in WSL2 on an NVIDIA GPU (CUDA) ? Windows-specific. ffmpeg is already cross-platform. The port must run the detector NATIVELY per OS (drop the WSL2 dependency): Linux+NVIDIA ? CUDA directly; macOS (Apple Silicon) ? NO CUDA ? Metal/MPS or a CPU path; Windows native (no WSL). THE CRUX is COMPUTE ON A TYPICAL ENTHUSIAST LAPTOP, which often has NO discrete GPU ? WASB is ~3.4× real-time on a laptop 2070 Super; on CPU it's far slower. So the gating question (the P0 spike): is CPU acceptable for a ~10-min clip? does Apple MPS work for the WASB/TrackNet weights? do we need a smaller/faster owned model (ties OWNED_MODEL_IMPLEMENTATION_PLAN)? a tiered "fast-if-GPU, slow-but-works-if-CPU"? This decides product feasibility ? the doc should call it the make-or-break unknown and propose measuring it.

SCOPE the doc with: vision + why; Phase 1 = the de-WSL native-compute port (per-OS); the app shell options (Tauri (small) vs Electron vs CLI+installer over the existing Python/FastAPI backend); the flow (detect USB/SD insert ? list videos ? on-device proxy + WASB + stitch ? export reels to a user folder, NEVER copy the originals); packaging + code-signing per OS; a §7 tracker (P0 compute-feasibility spike ? P1 de-WSL native detector ? P2 desktop app shell ? P3 packaging/installers); honest limits (feasibility hinges on the compute question; if CPU-only is too slow on typical laptops, the path may need a small owned model or a hybrid). This is OWNER-GATED implementation ? the scoping doc + (if greenlit) the P0 spike are the deliverable; do NOT build the app without explicit owner approval.

2. assistant (2026-06-18T07:43:10.966Z)

I'll start by following the repo-freshness and branching-safety conventions, then read the required context files. Let me kick these off in parallel.

3. user (2026-06-18T07:43:17.188Z)

---STATUS---

On branch feat/long-rally-lens

Your branch is up to date with 'origin/master'.

Untracked files:

(use "git add <file>..." to include in what will be committed)
artifacts/
docs/KHELSUTRA_PRODUCT_OVERVIEW.md

nothing added to commit but untracked files present (use "git add" to track)

---CURRENT BRANCH---

feat/long-rally-lens

---HEAD---

a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)

---ORIGIN/MASTER---

a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)

4. user (2026-06-18T07:43:17.307Z)

File content (284.4KB) exceeds maximum allowed size (256KB). Use offset and limit parameters to read specific portions of the file, or search for specific content instead of reading the whole file.

5. user (2026-06-18T07:43:17.315Z)

```
1      <!--
2      PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3
4      A "tracked project doc" is how this repo carries substantial design/project work: it can
be
5      ITERATED on, its progress TRACKED, and it is MARKED DONE against explicit deliverables,
then
6      archived. Standardizes the emergent pattern in docs/RALLY_QUALITY_RESEARCH.md §7 and
7      docs/GOLDEN_REGRESSION_FIXTURES.md.
8
9      Rules of thumb:
10     - §7 (the progress tracker) is the SOURCE OF TRUTH ? one row per independently-shippable
PR.
11     - Keep it HONEST: tag each claim [verified] (checked against code), [researched] (needs
sign-off),
12       or [design] (proposed). The doc reflects real shipped state, never aspirational.
13     - It POINTS to detail (code anchors, research artifacts), it does not duplicate it.
14     - Sections marked OPTIONAL are included only when relevant (e.g. Foundation/Threat-model
for
15       research- or security/privacy-heavy work). Delete sections that do not apply.
16     - Move to docs/archives/ once Status = DONE (per the archive policy: only terminal-state
work).
17     -->
18
19     # <Project / Plan Title>
20
21     > **Status:** `DRAFT` ? `<owner-gated? | in progress>` . **Owner:** `<name>` .
**Created:** `<YYYY-MM-DD>` . **Last updated:** `<YYYY-MM-DD>`
22     > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
23     > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
```

```

24     > **Relation to existing docs:** `<extends | supersedes | peer-of ?>`
25     > **Honesty note:** mark each claim `[verified]` / `[researched]` / `[design]`. Not
legal/financial advice where applicable.
26
27     ---
28
29     ## 0. TL;DR
30     <2?5 sentences: the goal, the approach, and the single most important caveat. Call out
any finding that is already actionable on its own.>
31
32     ## 1. Problem & goal
33     <Why now; the concrete user-facing outcome; what "good" looks like. Link the code/docs a
reader must read to get current.>
34
35     ## 2. Decisions locked
36     <Table ? capture answered questions so they are not relitigated.>
37
38     | # | Decision | Source / date | Implication |
39     |---|-----|-----|-----|
40     | D1 | ? | ? | ? |
41
42     ## 3. Foundation ? research / prior art *(OPTIONAL)*
43     <Include when the plan rests on external facts: cited research, legal/compliance,
benchmarks, prior-art survey. Mark researched-vs-verified; list any sign-off gates this
creates.>
44
45     ## 4. Design / architecture
46     <The actual plan. Data contracts, diagrams, and a REUSE MAP ? name the existing
modules/code this builds on so it is clear what is new vs reused.>
47
48     ## 5. Threat model / risk table *(OPTIONAL ? security / privacy / compliance work)*
49     <Who/what is protected vs NOT; residual risks stated plainly.>
50
51     ## 6. Honest limits ? what this does NOT do
52     <The false-confidence guardrail: what a green suite / a DONE status does NOT prove.>
53
54     ## 7. Deliverables & progress tracker    ? **source of truth**
55     Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
56
57     | ID | Deliverable | Depends on | Gated? | Status | PR |
58     |---|-----|-----|-----|-----|-----|
59     | P0 | ? | ? | No | ? | ? |
60
61     ## 8. Open questions ? owner / external *(split blocking-gates from nice-to-knows; name
who decides)*
62     1. ?
63
64     ## 9. Definition of done
65     <Explicit acceptance checklist. When all ticked ? flip `Status ? DONE` and archive.>
66     - [ ] ?
67
68     ## 10. References
69     **Internal code anchors `[verified]`:** `<paths>`. **Internal docs:** `<docs>`.
**External:** `<cited sources>`.
70
71     ---
72
73     ### Changelog
74     - `<YYYY-MM-DD>` ? `<what changed>`.
75

```

6. user (2026-06-18T07:43:17.331Z)

```

1     - [ROORA_LABELING_GUIDELINES.md](ROORA_LABELING_GUIDELINES.md) ? Latest (v8) vendor

```

labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven court-calibration / singles-doubles additions marked `? ADDED (fusion)`.

2 - RALLY_QUALITY_RESEARCH.md ? **Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a **shadow** signal), an ***adversarially-verified, cited external-research synthesis*** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a ***prioritized cheap?durable roadmap with falsifiable success criteria***, and (\$7) a ***tracked project plan*** ? work-item checklist, sequencing, the single quality ***number*** attributed to the effort, and a ***definition of done*** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.

3 - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc** (2026-06-18) for closing the remaining rally-detection accuracy gap ***per video**, before launch, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is ***detection***: (G1) recall (local & Gemini both miss ~half on hard clips, and **different** rallies ? complementary), (G2) precision (local over-fires), (G3) ***rally-END over-extension on the quick-return-after-point*** (shuttle still moving ? needs a **player-state** signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally\_gate` Gate A wiring, complementary fusion. ***Peer*** of `RALLY\_QUALITY\_RESEARCH.md` (execution tracker, not research); \$7 is the source of truth.

4 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint** (2026-06-17) of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

5 - EVALUATION.md ? How we measure and improve quality.

6 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with ***zero production regression*** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

7 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

8 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a ***signal-fusion framework*** for rally detection ? producers emit confidence-scored ***signals*** at three temporal ***scopes*** (frame detectors / window analyzers / session analyzers), a learned ***scorer*** weights them, and an auditable ***report-first reconciliation*** pass lints the decision against the evidence. The spine: ***no special-casing in the decision path*** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

9 - PERSONALIZATION_PLAN.md ? Design (owner-adopted ***hybrid***; Phase-0 landing) for the ***per-user personalization feature on a generalization-first baseline***: a thin per-user ***tuning*** layer gated by the honest small-N statistics (a ***min-N promotion floor*** + a ***structural-guard exemption***, never a service gate) + a ***contribution flywheel*** (free-tier videos pool into the owner-trained baseline; ***a tool that gets better the more you help it***). Records the ***locked owner decisions*** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

10

11 - GOLDEN_REGRESSION_FIXTURES.md ? **Design** (owner-gated, NOT started): ***video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with *no video/GPU/DB***. Two layers ? the post-detector freeze + ***dual gates*** (characterization + quality-floor) reusing the `backend/eval/` stack, and the ***SEALED-FIXTURES*** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited ***IN/US/EU legal memo*** (derived-data IP/privacy), a threat model,

```

and a progress tracker + definition-of-done. Extends
[GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md).
12
13     ## Practical / ongoing
14     - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
15     - Golden set docs (`GOLDEN_SET`)
16     - Evaluation harness and calibration drivers (in `backend/eval`)
17     - [GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md) ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `_golden_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See #175.
18
19     ## Conventions & templates
20     - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a tracked
project doc: status header · decisions-locked · progress tracker (???) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
21
22     ## Historical documents
23     Completed plans, old research, ADRs, and checkpoints live in [archives/](archives/).
24
25     - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? COMPLETED
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
26
27     See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.

```

7. assistant (2026-06-18T07:43:31.234Z)

Let me read the live state file (it's large, so I'll read the top) and the key strategy docs in parallel.

8. user (2026-06-18T07:43:32.360Z)

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9. user (2026-06-18T07:43:32.679Z)

```

1     # Platform Architecture ? Multi-User "Assistant Coach"
2
3     Status: ? Strategy / research ? for review. No code yet.
4     Date: 2026-06-07 · Owner: avidullu
5
6     > ? Superseded where it disagrees with
[OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
7     > (2026-06-09, authoritative). Specifically: (1) licensing is MIT/Apache-2.0/BSD
(not "Apache-2.0 only");
8     > (2) base model is Qwen3-VL (Apache) primary / InternVL3 (MIT) fallback ? Gemma-4
is AMBER/bench-only, not a
9     > cleared base; (3) the owned VLM is not the per-turn "conversational brain" ?
extraction (TAL head ? later VLM)
10    > feeds a per-video event store + text-to-SQL query layer ("extract once, query
many"); the VLM-answers-each-turn
11    > design is rejected. The §5A model/base/QA-harness detail below predates that and is
kept for context only.
12
13    Purpose: Map the path from today's single-user, local tool to a multi-tenant

```

```

14 "assistant coach" platform**: users upload videos, get highlights **stored and
15 browsable by history**, and choose **where their data lives** (Google Drive / cloud
16 bucket / local) and **where the compute runs** (our GPU / their VM). The organizing
17 principle is to **decouple the layers** ? identity, ingestion, storage, orchestration,
18 execution/compute, the detection+highlight pipeline, and data-serving ? behind small
19 **pluggable interfaces**, reusing the registry pattern the codebase already uses for
20 segmenters/providers/validators/sports/annotations.
21
22 **Product framing (the north star *and* the v1 line).** This is an **assistant coach**,
23 not just a clipper ? but the strategy is **phased and unambiguous about scope**:
24 - **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-
user
25 history ? a coach/user correction loop that feeds the golden set** ("assistant-coach
26 *lite"). Essentially today's pipeline made multi-user ? this is the real v1
deliverable.
27 - **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the
28 **owned video-QA model** ($5A; $6 P4?P5).
29 - **North-star vision (strictly future ? not* v1):** **gait / biomechanics, per-athlete
30 identifying templates, and deep coaching feedback under a supervising coach** ($6 P6).
31 Inspiring, but explicitly deferred ? and the part that raises the **biometric**
privacy
32 bar ($7).
33
34 The buyer is the **academy/coach** (beachhead in
35 [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two
36 consequences ripple through the design: (1) the data model keeps an **optional coach ?
37 athlete overlay over a generic core** ($4.1) ? not baked in; (2) gait/biometrics is
38 **special-category data**, gated and deferred ($7).
39
40 > **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
41 > Spideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
42 > (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
43 > your storage, your compute" is a genuine differentiator for privacy-sensitive
44 > academies ? and SwingVision proves user-side compute is commercially viable. All four
45 > share the same **teams ? athletes ? clips** data model, which we adopt ($4).
46
47 > Scope: this is a **direction-setting doc**. It does not start implementation; the
48 > first infra slice (an async job model) is the on-hold #16 in
49 > [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints
come
50 > from [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
51 > foundations) and `archives/MONETIZATION_AUDIT.md`.
52
53 ## 1. Current state & gaps (what we're evolving from)
54
55 | Concern | Today | Gap for multi-user |
56 |---|---|---|
57 | Identity / tenancy | none | no users, orgs, auth, or isolation |
58 | Storage | local `filepath` string; output to `output_dir` | no abstraction; can't BYO
Drive/bucket; raw video must be on the box |
59 | Data model | `videos` keyed by file **MD5**, `play_segments`, `events`,
`candidate_segments` (SQLite) | no owner/athlete link; **highlights aren't a first-class
persisted entity** (compile writes a file and returns a path); no per-user history |
60 | Execution | **async job model shipped** ? `/api/process` returns **202 + job_id**, a
single-worker `ThreadPoolExecutor` drains the `jobs` table, `GET /api/jobs/{id}` polls status |
single-tenant + local-only; can't dispatch to a user's VM/GPU yet |
61 | Pipeline | **already pluggable** (segmenter/provider/validator/sport/annotation
registries) | reuse as-is ? this layer is the model for the rest |
62 | API | `/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth | needs
auth, per-tenant scoping, async job + history endpoints |
63
64 Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version`
65 migration ladder, **v5** ? `jobs` table + per-user `user_models` already landed),
66 `backend/main.py` (endpoints + the async job worker ? `/api/process` ? 202 + job_id,
67 `GET /api/jobs/{id}`), the registry pattern in

```

```

68     `backend/pipeline/segmenters/base.py::SegmenterRegistry` and
69     `backend/providers/base.py`.
70
71     ## 2. Target layered architecture
72
73     Decouple into layers with a clean **control-plane vs data-plane** split ? the
74     orchestrator coordinates *metadata and jobs*; the heavy *bytes and GPU work* can live
75     elsewhere (even in the user's own cloud).
76
77     ...
78
79     CONTROL    ? Presentation / API (auth, per-tenant) ?
80     PLANE      ? Identity & Tenancy (org?coach?athlete?user) ?
81               ? Orchestration / Jobs (submit?status?result) ?
82               ?????????????????????????????????????????????????????????????
83               ? dispatch ? metadata
84               ?????????????????????????????????????????????????????????????
85     DATA      ? Execution / Compute ? ? Data / Serving ?
86     PLANE      ? (local | BYO-VM | ? ? (per-tenant DB: ?
87               ? hosted GPU) ? ? videos, segments, ?
88               ? runs the Pipeline: ? ? highlights, jobs, ?
89               ? detect ? highlight ? ? history) ?
90               ?????????????????????????????????????????????????????????????
91               ? read input / write artifacts
92               ?????????????????????????????????????????????????????????????
93               ? Storage (local upload | Google Drive | S3 | ?
94               ? GCS | Azure) ? raw video + highlight outputs ?
95               ?????????????????????????????????????????????????????????????
96     ...
97
98     The two **new pluggable seams** are **Storage** and **Compute/Execution**; everything
99     else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A
100    **model/training layer** ? the owned generative video-QA model ($5A) ? sits atop these,
101    fed by Storage (training corpus), Data (labels), and Compute (training + inference), and
102    is delivered through the existing **AI-provider registry with a paid fallback**.
103
104    ## 3. Pluggable abstractions (extend the existing registry pattern)
105
106    The codebase already proves the model: an `ABC` + a `Registry` singleton with
107    `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with
108    the same shape so they're plug-and-play and testable in isolation.
109
110    ### 3.1 `StorageBackend` registry
111    Replace the bare `video_path: str` that flows through the pipeline with a
112    **`StorageRef`** (a backend id + a key/URI), resolved by a backend.
113
114    ```python
115    class StorageBackend(ABC):
116        name: str
117        def open_read(self, ref: StorageRef) -> BinaryIO: ... # stream large video
118        def download_to_local(self, ref: StorageRef, dst: str): ... # for ffmpeg/cv2/WSL
119        def upload(self, local_path: str, ref: StorageRef): ... # persist
120        def highlight/artifacts
121        def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
122        def stat(self, ref: StorageRef) -> StorageStat: ...
123        def list(self, prefix: StorageRef) -> list[StorageRef]: ...
124
125    ...
126
127    - **Recommended implementation:** build the backend over **`fsspec`** (one matrix ?
128    local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`,
129    Dropbox via `dropboxdrivefs`), reaching for **`smart_open`** as the robust large-MP4
130    byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as
131    add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
132
133    - **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**:
134    the tenant registers their own bucket/Drive; raw video **never leaves their account**.

```

131 - **Auth** ? never persist users' long-lived keys.** For buckets, have the user create an
132 IAM role you **AssumeRole** into, scoped to one bucket/prefix, and mint **presigned**
133 URLs** so raw video moves **browser** ? the user's bucket directly, never through our
134 servers**. For Drive, OAuth with the narrow **drive.file** scope; for academies, a
135 **service account with domain-wide delegation**. Store any per-tenant creds encrypted.
136 - **Why it matters for the pipeline**:** ffmpeg/OpenCV/WSL need a **local file**, while
the
137 LLM provider can take a **signed URL**. The backend exposes both **download_to_local**
138 (for compute) and **signed_url** (for the provider/browser), so the pipeline stops
caring
139 **where** the bytes live.
140 - **Migration path**:** introduce **StorageRef** alongside the current local path; a **local**
141 backend wraps today's behavior so nothing breaks; adapters land incrementally.
142
143 **### 3.2 ComputeBackend / ExecutionBackend registry**
144 where the pipeline actually runs for a given job.
145
146 - **local** ? in-process / local subprocess (today's behavior; dev + self-host).
147 - **hosted** ? our GPU VM pool (the SaaS default).
148 - **byo_worker** ? the tenant runs a **pull-based worker** (a small agent) on their
own
149 VM/GPU. **Critical security property**:** the worker **pulls** jobs scoped to its tenant
150 from the control plane, pulls input from the tenant's **own** storage, runs the
151 pipeline, and writes artifacts + results back ? so **the orchestrator never holds the**
152 **user's cloud credentials** and needs **no inbound access** to their network. This
153 mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s
154 outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most**
155 **credential-safe v1** for BYO-compute.
156 - For users who want the platform to **provision GPUs** in their **own** cloud account on
157 demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot**
158 (BYOC: everything launches in the user's accounts/VPCs; creds stay with them).
159 **Modal** is the managed fallback for users wanting zero infra (accepting that raw
160 video then flows through Modal).
161 - **Operational / support burden** (a real adoption risk).** For the primary beachhead
?
162 Indian badminton **academies/coaches** ? asking them to **run and maintain a worker** on
163 their own VM/GPU* is a meaningful ongoing load (updates, monitoring, firewall/NAT, the
164 "it stopped after our IT change" ticket). So: **default academy users to hosted**,
and
165 treat **byo_worker/SkyPilot** as a **power-user / enterprise / privacy-sensitive tier**,
166 not the academy default. If/when BYO ships, plan for **auto-updating, self-healing**
167 **agents + health heartbeats + clear worker status** in the UI**, and budget the support
cost.
168 - **Middle tier to consider** ? "zero-infra BYO":** a managed-but-isolated path where the
169 **storage** stays in the user's bucket (signed URLs) while compute runs on **our**
GPUs
170 in a **strictly isolated, per-tenant, ephemeral sandbox** (nothing persists our side).
171 This gives the BYO **privacy** benefit (raw video never persists on our infra)
without
172 the user running a worker ? a pragmatic default between full-hosted and full-BYO.
173
174 The pipeline code is unchanged; only **where it's invoked** and **how it gets bytes** vary.
175
176 **### 3.3 Reuse, don't reinvent**
177 The existing **segmenter** / **provider** / **validator** / **sport** / **annotation** registries
178 are the proven precedent and stay as the **pipeline** layer. The **annotation** registry +
179 **candidate_segments** table are also the natural seam for the **coach-correction / golden**
180 **-set** loop in the assistant-coach phase (\$6, P4).
181
182 **## 4. Multi-tenant data model & serving layer**
183
184 **### 4.1 Entities** ? generic core, with the coach model as an optional overlay
185 The foundational schema is deliberately **domain-neutral** so we don't overfit to the
186 coach mindset ? the coach/athlete framing is a **market-fit hypothesis** that will
iterate**


```

187 (see C13 / PMF), not a load-bearing assumption. **Generic core:**
188 ```
189 User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
190      ???< PlaySegment ??< Event
191 Video / Highlight ??< Job (processing record: status, compute backend, timings)
192 ```
193 - **Generic & re-targetable:** a `VideoCollection` groups a user's videos (a season, an
194   event, a "folder") with no assumption of a coach or academy ? a power user, a coach,
or a
195   parent are all just `User`s with collections.
196 - **Highlights are first-class** (today compile only writes a file): a `highlights`
table
197   (segment ids + output `StorageRef` + params + created_at) powers "see past videos and
198   highlights by history," and **`HighlightMetadata`** carries the **rich semantic tags**
199   (shot type, rally outcome, etc. ? see §6 metadata stage) plus titles / ratings /
notes.
200 - **`video_id` stays the content MD5** (dedup within a tenant), scoped by `tenant_id`.
201 - **Coach overlay (optional, additive):** for the academy/coach GTM, layer
202   `Org/Academy ? Coach ? Athlete` *on top* (e.g. a `Membership`/role table + an
`athlete_id`
203   tag on collections/videos) **without** baking it into the core tables ? so the schema
204   stays usable for solo power users and the coach model can evolve, or be dropped, as
PMF
205   dictates.
206 - **History/serving API:** list collections/videos/highlights per user (and, with the
207   overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g.
shot
208   type / outcome); re-stitch from stored segments without reprocessing.
209
210 ### 4.2 Tenancy isolation ? options & recommendation
211 | Model | Isolation | Ops cost | Best for |
212 |---|---|---|---|
213 | **Shared DB + `tenant_id`** (row-level, ideally Postgres **RLS**) | logical | low |
**Hosted SaaS, many small tenants** ? *recommended default* |
214 | Schema-per-tenant | medium | medium | tens?hundreds of mid tenants |
215 | **DB-per-tenant** (incl. **SQLite-per-tenant**) | strong/physical | higher at scale |
**self-hosted single-academy**, or high-isolation enterprise |
216
217 **Recommendation:** for the hosted product, **migrate SQLite ? Postgres with shared-DB +
218 `tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted /
219 single-academy** deployment (it reuses today's code almost verbatim). Either way,
220 **abstract DB access behind a repository layer now** so the engine is swappable, and
221 graduate the `PRAGMA user_version` migration discipline to **Alembic** under Postgres.
222
223 ## 5. Orchestration / async jobs
224
225 This is the serving layer that makes multi-user real. It is the **multi-user superset of
226 [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ?
build
227 that first, single-tenant, then generalize.
228
229 - **Job record** (`jobs` table): `id, tenant_id, video_id, type (index|compile|analyze),`
230   status (queued|running|done|failed), progress, compute_backend, result_ref, error,
231   timestamps`. `/api/process` returns **202 + job_id**; `/api/jobs/{id}` polls.
232 - **Idempotency** keyed on `(tenant_id, video_id, pipeline_version)` ? reuse the
233   content-hash discipline already in `compute_video_id`.
234 - **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs
235   for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline,
236   writes artifacts + DB rows. **Observability reports** (already per-video) become the
237   job artifact surfaced in history.
238 - **Queue/worker backend ? a decision to prototype in #16, *not* a default here.** The
239   **requirements** are firm ? **idempotency**, **resumability** (a 40-min GPU job mustn't
240   restart on a worker blip), **progress states** ? which rule out fire-and-forget
Celery/RQ
241   for the long jobs. The *choice* should be made by prototyping in the

```

```

242     [ `DEFERRED_HARDENING_PLAN.md` ] (DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:
243     - **Start simplest:** **Arq** (async-native, natural FastAPI fit) or even a **plain
244       Postgres-backed queue + worker ** ? likely sufficient for the *first* single-tenant
245       async slice, with the lowest operational complexity and the easiest debugging.
246     - **Durable-execution upgrade if/when needed:** **Hatchet** (Postgres-backed, per-step
247       retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
248       because it adds **no new datastore** once we're on Postgres, but it *is* a new
249       dependency + operational/debugging surface; adopt only when long-job resumability/
250       observability demands it.
251     - **Heaviest:** **Temporal** ? most mature durable workflows, operationally heavy;
252       reserve for genuinely complex multi-stage pipelines at scale.
253
254     Record the decision criteria in #16: operational complexity, maturity, **debugging
255     experience for long GPU jobs **, and whether durability is worth a new dependency.
256     **Bias: start with the simplest thing that gives retries + status; graduate only on
257     evidence. **
258
259     ## 5A. Owned generative model (the video-QA assistant) & the data flywheel
260
261     The north-star capability: an **owned, self-hostable generative video model** ?
262     fine-tuned on sports video (including **consented, user-contributed** footage) ? that
263     **answers questions about rallies, shots, and technique**. This is the conversational
264     brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
265     is introduced **behind the same abstraction, with the paid API as an always-ready
266     fallback **.
267
268     > **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
269     > assets are ** (a) the consented, badminton-specific video?QA dataset** and ** (b) the
270     > eval harness **. Build and message *those* ? not "we own an LLM."
271
272     This conversational surface (NL queries + coach-mode recommendations) must be
**localized
273     from the start ** for the Bangalore beachhead ? the query *and its answer* in the user's
274     language. The locale foundation and the templated-answer approach are specified in
275     [ `I18N_PLAN.md` ] (I18N_PLAN.md) (phase P4 ties to this section).
276
277     > **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings
in
278     > full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
279     > serving / fallback ** only ? never as a *training* data source.** Harvesting their
outputs
280     > to train our owned model would breach their "no competing model" terms (a
terms/copyright
281     > violation), so it is prohibited in both engineering and legal process. **Training
teachers
282     > are open, Apache-2.0 models only ** (Qwen-VL / InternVL / Gemma 4). Detail in the
283     > distillation constraint below.
284
285     ### Core principle: AI is a pluggable provider with a paid fallback
286     The codebase already abstracts the cloud AI behind `provider_registry` /
287     `AIVideoProvider` (`backend/providers/`). Make that a **first-class platform
principle **:
288     - Every generative capability routes through the **provider registry**.
289     - The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
290       Vertex AI ** stays registered as the **fallback** (failover on error/low-confidence,
291       or as the quality ceiling/teacher reference).
292     - This lets us **ship today on the known paid stack** and **swap in the owned model when
293       its quality clears a measured bar ** ? with zero architectural churn and the paid path
294       always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
295       owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)
296
297     ### The model / training layer (new sub-layer, fed by the scaffolding)
298     ```
299     Data curation      Training      Eval      Serving
300     (consented user vid (fine-tune an open, (extend the existing (self-host inference

```

```

301 + golden set +      commercially-usable  eval harness to QA  as a provider /
302 rally/segment/event base VLM via      quality + the      ComputeBackend;
303 annotations ?      LoRA/QLoRA)          regression gate)   paid fallback)
304 video-QA pairs)
305 ```
306 This layer **reuses every other layer**: **Storage** holds the training corpus;
307 **Data** supplies labels (the `annotation` registry + `candidate_segments` +
308 golden set are already the labeling seam); **Compute/Orchestration** runs training and
309 inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the
310 **eval harness** already in `backend/eval/` extends to grade QA quality. **In other
311 words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ?
312 which is why scaffolding comes first (see §6).
313
314 ### Base model & training (Apache-2.0 only ? licensing is decisive)
315 **Recommended near-term base: `Qwen2.5-VL-7B-Instruct`** ? Apache-2.0, native long-video
316 (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path
317 **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean
318 Google-stack alternative to evaluate.
319 - ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private,
320 gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL
321 is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
322 - ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative
323 clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays
324 permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch
to
325 Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?"
326 model name + "Built with Llama" badge and a 700M-MAU ceiling. **gpt-oss** ?
Apache-2.0
327 but **text-only**, so usable only as an optional text-reasoning layer, never the video
328 encoder.
329 - **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and
330 avoids catastrophic forgetting of general video skills. Build the dataset by
converting
331 the existing rally/shot/segment annotations + golden set into templated **clip?QA
332 pairs**, expand synthetically with **open-model teachers only** (see constraint
below),
333 and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs
?
334 **5k?20k** curated is a realistic band for this narrow domain.
335 - **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is
a
336 comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you
have
337 sustained video-QA volume **+++ a defensible consented dataset **+++ real per-call API
338 cost pain ? below that, the paid API is cheaper than a standing GPU and the second
stack
339 is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**
340
341 Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.
342
343 ### Training backend ? managed service vs self-managed (the fine-tune "forge")
344 *Where* you run the LoRA/RL fine-tune. Weigh on **video support, weight
export/ownership,
345 smallest servable model, RL support, no-train DPA, and infra burden**.
346
347 **Decisive finding (researched 2026-06):** managed fine-tuning services are
348 **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them
349 as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native*
350 video (+audio) training**. So for true video-QA, a self-managed framework is the
351 strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to
352 serve) and keep full ownership. Managed services trade some of that for zero infra.
353
354 | Option | Type | Video / VLM support | Smallest VL | RL | Export weights | Best for |
355 |---|---|---|---|---|---|---|
356 | **ms-swift** (ModelScope) | Self-managed (rented/own GPU) | **Native video+audio**;

```

Qwen3-VL/Omni, 300+ MLLMs | any incl **7B** | ? DPO/GRPO | ? full ownership | *** video-QA; slots into BYO-compute (§3.2)** |

357 | **LLaMA-Factory** | Self-managed | Qwen3-VL + 100+ VLMs; video datasets | any incl 7B | ? | ? | popular, well-documented |

358 | **Unsloth** | Self-managed (1-GPU optimized) | Qwen3-VL LoRA (vision+lang; frames) | any incl 7B | ? GRPO | ? | cheapest first runs |

359 | **Tinker** (Thinking Machines) | Managed API (low-level) | Qwen3-VL **30B-A3B+ only**;
image input (video **unverified**) | 30B-A3B MoE | ? PPO/GRPO, custom loss | ? adapters + ckpt;
no-train ToS | zero-infra RL/research at MoE scale |

360 | **Together AI** | Managed (tune + serve) | Qwen3-VL / Gemma-3 / Llama-4; image data | verify | partial | ? adapters downloadable | managed Qwen3-VL + serving |

361 | **Fireworks AI** | Managed (tune + serve) | Full Qwen2.5-VL (3B?72B SFT; LoRA@32B + LoRA-upload) | **7B** | partial | ?? verify export (lock-in risk) | managed *small*-VL + serving |

362 | **Predibase** (Rubrik) | Managed *** VPC self-host** | VLM instruct-tune; **LoRAX** multi-adapter | verify | ? (VPC) | ? **RFT/GRPO** flagship | RL "creativity" + multi-adapter + privacy |

363 | ~~OpenAI GPT-4o / Gemini tuning~~ | Closed | ? | ? | ? | ? no export | **non-fit ? violates "own the model"*** |

364

365 * (Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full
366 control but are only worth it if you're already on that stack.)*

367

368 **Recommendation:**

369 - **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory /
370 Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud):

native

371 video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the
372 **ComputeBackend / BYO-compute** seam (§3.2). **Likely a better fit than Tinker** for
373 the video requirement *and* serving economics.

374 - **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving)
375 or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support
and
376 adapter export.

377 - **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase**
378 (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-
379 level RL primitives).

380 - **Tinker** remains a strong zero-infra RL-research option with clean weight export ?
just

381 not the cheapest path to a *small video* model (its VL floor is 30B-A3B).

382 - **Portability is the hedge:** you own the **data + recipe + exported weights**, so the
383 backend is swappable ? same ethos as the provider/registry pattern. Don't get locked
in.

384

385 **Verify before committing (any backend):** (1) **native video / frame-sequence**
386 ingestion + max frames per example; (2) **weight export** + a **commercial DPA**
covering

387 consented, minors-excluded video (biometric, §7); (3) the **inference plan** for the
size

388 you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).

389

390 ### Training harness ? make it agent-runnable (nanochat-style operability)

391 The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled
video

392 and get a trustworthy ship / no-ship verdict, unattended.** Karpathy's **nanochat** is
the

393 reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs
394 tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card**
395 score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its
396 five properties are exactly what make training agent-runnable, and we copy them:
397 ** (1) one command, end-to-end; (2) one complexity dial (sane auto-derived
defaults);

398 ** (3) an automatic **report card** ? a deterministic pass/fail vs a target (the crux);

399 ** (4) reproducible + deterministic + cheap; (5) minimal/readable.

400

```

401  **Engine (don't reinvent):** `ms-swift` already delivers this for VLMs ? CLI-driven,
402  **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via
403  `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF `nanoVLM` ?
404  ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)
405
406  **The harness = thin glue over components we already have** ? one declarative,
407  agent-invokable job (call it `train_speedrun`):
408  1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry
+
409  `candidate_segments` + golden set ? the labeling seam in `backend/`).
410  2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an
orchestration
411  job (§5 / DEFERRED #16): one config, idempotent, logged.
412  3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness
to
413  **QA quality**; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
414  4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays)
**only
415  if it beats the bar**.
416
417  So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register
418  provider**, behind one command an agent or CI runs. nanochat is the inspiration for the
419  wrapper; ms-swift is the engine; the **golden QA eval is the report card**.
420
421  > **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA
eval
422  > set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic
and
423  > trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
424  > optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)
425
426  ### ?? Teacher / distillation constraint (the single biggest legal landmine)
427  You **cannot** train the owned model on outputs from the paid frontier APIs ? a general
428  video-QA assistant is squarely a "competing model": **Gemini** ("may not use the
429  Services to develop models that compete"), **OpenAI** ("may not use Output to develop
430  models that compete"), **Anthropic** (bars using Outputs to train competing models). Two
431  safe lanes: **(1)** generate synthetic QA with **open-model teachers only** (Qwen-VL /
432  InternVL / Gemma 4 ? Apache-clean); **(2)** keep the paid API strictly as a **runtime
433  fallback** ? serving answers is fine, *harvesting its outputs to train the model that
434  replaces it is not*. Write a hard "no training on competitor-API outputs" rule into
435  eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:
436  the teacher is *open models*, not the paid APIs.)
437
438  ### Data flywheel, personalization & consent tiers
439  Two complementary loops, each with its **own** opt-in (training consent is separate from
440  processing consent ? §7):
441
442  - **Global model (shared ? raises the floor).** Train the base owned model on an
443  **explicitly-contributed, licensed** corpus ? *not* on whatever sits in a tenant's
bucket
444  (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
445  better shared model ? better highlights/answers ? more reason to contribute.* Higher
446  consent bar (pooled data, biometric/minor care ? §7); **minors excluded by default**.
447
448  - **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under
exploration.**
449  A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
450  consented videos*** on top of the shared base, served via **multi-LoRA** (LoRAX /
451  Together / Predibase). Why it's compelling:
452  - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
453  styles ? materially better detection/highlights for them.
454  - **Cleanest consent story:** "improve *your* detection using *your* data" is
far
455  lower-friction and lower-risk than contributing to a global model ? the data
improves

```

```

456     **only the user's own adapter**, with **no cross-user mixing**, sidestepping much of
the
457     pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your
video,
458     your storage, your compute ? and now **your model**.* With BYO-compute the adapter
can be
459     trained/served on the user's own GPU and **never leave their boundary**, **deletion
=
460     drop the adapter** (clean erasure).
461     - **Open questions (still a hunch):** cold-start (need enough of the user's labeled
data
462     before a personal adapter beats the base ? fall back to the shared model meanwhile);
a
463     **per-user held-out eval** to prove the adapter helps (don't ship a regression);
base +
464     optional global-adapter + per-user-adapter stacking; per-user training cost at
scale.
465     Worth a spike once the harness + golden eval exist.
466
467     **Two consent tiers, then:** *(a)* "improve **my own** detection with **my** data"
468     (per-user, low friction) and *(b)* "contribute to the **shared** model" (global,
higher
469     bar). **Default for both is off.**
470
471     ## 6. Phased roadmap
472
473     | Phase | Deliverable | Build / reuse |
474     |---|---|---|
475     | **P0** | ? **DONE ? Async job model, single-tenant** (DEFERRED #16) | `jobs` table via
`user_version` ladder; single-worker `ThreadPoolExecutor`; `/api/process` ? 202 + job_id; `GET
/api/jobs/{id}` status |
476     | **P1** | **Identity + multi-tenant data model + upload + highlight persistence +
history** | auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS;
`highlights` table; repository layer; history API + UI |
477     | **P2** | **`StorageBackend` registry + BYO-storage** | new ABC+registry;
`local`?`gdrive`/`s3`/`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
|
478     | **P3** | **`ComputeBackend` registry + BYO-compute** | `local`/`hosted`/`byo_worker`;
pull-based worker agent; control/data-plane split |
479     | **P4** | **Rich rally/shot semantic metadata & tagging** *(the next value stage after
solid rally detection + highlights)* | shot type (smash / backhand drive / net), rally
**outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ?
query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden
set via the `annotation` registry + `candidate_segments` |
480     | **P5** | **Owned generative video-QA model** ($5A) | global flywheel **+ per-user
adapters** ; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as
`owned_vlm` **provider with Gemini paid fallback** ; extend the eval harness to QA quality |
481     | **P6** *(visionary)* | **Gait / biomechanics under a supervising coach** |
deliberately **back-burner** ; technique feedback; highest Art. 9 bar ? gate behind explicit
biometric + parental consent ($7) |
482
483     P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3
484     deliver the "plug-and-play storage/compute" ask. **P4 (rich semantic metadata) is the
485     immediate next value stage once rally detection + highlights are solid** ? e.g.
**"rallies
486     that ended in a smash or backhand drive,"* **"rallies where the court was open but the
487     player made an unforced error."* **P5** owns the model; **P6 (gait) is the visionary
488     horizon, intentionally deferred.**
489     **Sequencing principle (owner directive):** make the scaffolding (P0?P4) **rock-solid
490     first**, keep the **known paid stack as an always-ready fallback**, and only then shift
491     focus to **owned-model quality** (P5). The provider registry keeps the paid path one
492     config flip away.
493
494     ## 7. Privacy & legal (gait raises the bar ? design for it now)
495

```

```

496 - **The load-bearing nuance:** biometric data is special-category **only when used to
497 *uniquely identify* a person** (GDPR Art. 9). **Highlight/shot detection on players a
498 coach already knows is general analytics ? generally *not* Art. 9.** But
499 **gait/biomechanics that builds a per-athlete identifying template tips into Art. 9**
500 and needs **explicit, granular consent** referencing biometric data. Engineer gait
501 with that boundary (consent gate, purpose limitation, no cross-account
identification);
502 decide whether it runs **locally/on-tenant only** (strong posture) vs our cloud. (EDPB
503 Guidelines 3/2019 on video devices apply.)
504 - **Minors (e.g. academies).** Parental/guardian consent and safeguarding are first-
order,
505 not an afterthought. Model consent explicitly per relationship (user?subject,
coach?athlete
506 where the overlay applies, parent?minor).
507 - **BYO-storage is the central privacy/liability lever:** if raw video stays in the
508 tenant's own bucket/Drive and we process via signed URL or their compute, we minimize
509 what we hold ? shrinking breach surface, residency, and erasure obligations.
510 - **Paid-LLM no-train (C4):** use **Gemini via Vertex AI** (contractual no-training),
511 **not the consumer/free tier**; set short/zero retention, document it in the DPA, and
512 surface "your footage is never used to train AI" as a trust feature for academies. For
513 the gait phase, prefer keeping biometric inference off third-party APIs entirely.
514 - **Training consent is a distinct, higher bar (for the owned model, §5A).** Training on
515 user video needs a **separate, explicit, opt-in training licence** ? not bundled with
516 processing consent ? and it **compounds** the biometric (Art. 9) and minor-consent
517 issues above. Train **only on explicitly contributed/licensed video**; never train on
518 what merely passes through a tenant's bucket. **Exclude minors' video from the
training
519 pool by default**, keep the training corpus a **separate, firewalled, opt-in sub-
pool**,
520 and log **per-clip consent/provenance** for revocation + audit. Incentives must not
make
521 consent non-free (the product stays fully usable without contributing). And the
522 **teacher-distillation rule** (§5A): never train on paid-LLM outputs. **Two consent
tiers
523 (§5A):** (a) **"improve **my own** detection with **my** data" ? a per-user adapter
trained
524 only on the user's own video (lowest friction; no cross-user mixing; deletion = drop
the
525 adapter); (b) **"contribute to the **shared** model" ? pooled corpus, higher bar.
Default
526 for both is **off**.
527 - **Right to erasure / retention:** per-tenant, per-athlete delete must cascade across
DB +
528 storage artifacts (the `ON DELETE CASCADE` discipline already in `database.py` extends
here).
529
530 ## 8. Key decisions for the owner (the forks)
531
532 1. **Hosted-first vs BYO-first.** Recommended: **hosted control-plane first** (P0?P1)
with
533 **BYO-storage soon after** (P2); **BYO-compute later** (P3) as a power-
user/enterprise
534 option ? it's the highest-complexity, highest-security item.
535 2. **Database now or later.** Move to **Postgres at P1** for the hosted path, or stay
536 SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access
537 regardless.)
538 3. **Auth: build vs buy** (Auth0 / Clerk / Supabase Auth / Cognito).
539 4. **Orchestration tech** (see research recommendation) ? bias to the simplest that
gives
540 retries + status.
541 5. **Gait processing location** ? local/on-tenant (privacy-max) vs cloud (capability-
max).
542 6. **How far to push BYO-compute** given its security/ops complexity.
543 7. **Owned-model base + licensing (§5A).** Which commercial-OK video VLM to build on
544 (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial

```

545 derivative is permitted ? see COMMERCIALIZATION C14.

546 8. ****Training-data sourcing & consent.**** Opt-in contribution terms, incentives, and the

547 teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).

548 9. ****When to start the owned model**** ? strictly after the scaffolding + flywheel are

549 solid (the explicit directive), with the paid provider as the standing fallback.

550

551 ****Risks:**** scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS +

552 tests); BYO-compute security; ****biometric/minor liability**** (get this right before any

553 gait feature ships); ****training on user video without proper opt-in/licence****, and

554 ****base-model / teacher-distillation licence violations****; cost of hosted GPU (training

555 ***and* inference)**; ****premature model-building**** before data/eval are ready (mitigated by

556 keeping the paid fallback and gating P5 behind the flywheel).

557

558 **## 9. Sources**

559

560 External best-practices research (2026-06-07) backing the recommendations above. Each

561 underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and

562 should be re-validated before committing budget.

563

564 ****Storage abstraction:**** [fsspec](https://fsspec.com/) ·

[cloudpathlib](https://github.com/drivendataorg/cloudpathlib) ·

[smart_open](https://github.com/piskvorky/smart_open) · [Apache

Libcloud](https://libcloud.apache.org/) · [gdrive-fsspec](https://github.com/fsspec/gdrive-

fsspec) · [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) · [AWS S3 presigned

URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) · [token-

based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) ·

[signed URLs across clouds](https://medium.com/@shaikhsharwan49/signed-urls-in-aws-gcp-and-azure-

a-beginner-friendly-guide-e2e549425865)

565

566 ****Multi-tenancy:**** [Bytebase: multi-tenant DB

patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/)

· [Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-

architecture) · [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-

tenancy-in-postgres) · [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-

tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)

567

568 ****BYO-compute / orchestration:**** [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) ·

[Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) ·

[GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-

runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-

controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control

plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS

Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank:

Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-

deployment) · [DevPro: Python task queues

2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-

comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-

task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) ·

[Hatchet](https://github.com/hatchet-dev/hatchet)

569

570 ****Privacy / legal:**** [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local:

biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) ·

[Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) ·

[EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb

_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video

guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-

final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-

guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-

about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/site

s/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention

compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss:

enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-

enterprise-llm-platform-guide)

571

572 ****Comparable products:**** [Veo](https://www.veo.com/) · [Veo Help

Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

573

574 **Training backends & harness (\$5A):** [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)

575

576 **Owned model & licensing (\$5A):** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) · [gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs & training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlmtrl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)

577

578 ## See also

579 - [``COMMERCIALIZATION.md``](COMMERCIALIZATION.md) ? C12 (SaaS foundations), C4 (paid-LLM), economics.
580 - [``DEFERRED\_HARDENING\_PLAN.md``](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the P0 infra slice), #7 packaging.
581 - [``PMF\_MARKET\_RESEARCH.md``](PMF_MARKET_RESEARCH.md) ? academy/coach beachhead (who this serves).
582 - [``CODE\_MAP.md``](CODE_MAP.md) ? current architecture + the registry extension points.
583 - [``ROADMAP.md``](ROADMAP.md) ? offline self-improving rally detection (the pipeline's own trajectory).

584

10. assistant (2026-06-18T07:43:42.949Z)

Now let me read the live state and locate the key pipeline anchors (WASB/WSL2 detector, stitch, generate_highlights) so the doc is code-grounded.

11. user (2026-06-18T07:43:44.899Z)

File content (27697 tokens) exceeds maximum allowed tokens (25000). Use offset and limit parameters to read specific portions of the file, or search for specific content instead of reading the whole file.

12. user (2026-06-18T07:43:45.927Z)

```
Found 31 files
tools\generate_highlights.py
tests\test_experiment.py
backend\results.py
backend\eval\experiment.py
backend\config\models.py
tests\test_tracknet_runner.py
tests\test_fusion_hybrid.py
backend\pipeline\segmenters\trajectory_hybrid.py
backend\pipeline\detectors\tracknet_runner.py
tests\test_tracknet_hybrid.py
backend\pipeline\detectors\wasb_infer.py
backend\pipeline\detectors\wasb_runner.py
tests\test_wasb_cache.py
tests\test_wsl_tilde_quoting.py
tests\test_wasb_runner.py
backend\pipeline\detectors\base.py
backend\utils\run_telemetry.py
tests\test_rally_gate.py
tools\cleanup_caches.py
tests\test_cleanup_caches.py
backend\pipeline\segmenters\wasb_hybrid.py
backend\pipeline\segmenters\tracknet_hybrid.py
tests\test_reliability_hardening.py
backend\pipeline\detectors\__init__.py
tests\test_detector_keying.py
tests\test_eval_partial_labels.py
run_all_tests.py
tests\test_calibrate_local.py
tests\test_distill_local.py
tests\test_export_wasb_segments.py
tests\test_calibrate_wasb.py
```

13. user (2026-06-18T07:43:51.079Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-18 (? **R1 MILESTONE LAUNCHED ? #204 MERGED** `67b6927`; flip
re-verified at **n=13**, owner-approved). Owner: "verify #204 still stands at n=13, merge if
findings stand" ? it stands (stronger), verified 3 ways, merged. **Local-windowing defaults now
ON in production: cap=6 + R4** (`max_window_duration 6`, `window_inactivity_end 1.5`,
`inactivity_rally_thresh 0.20`, `inactivity_min_density 0.30`). W2 / W1.5 hard-cap / R5 stay
default-OFF.
13     - ? **n=13 re-verify (2 new Boxhill clips GX010137/GX010141):** baseline?milestone toIF1
**?+0.222, 12/0/1, p=0.0005** (stronger than n=11's 10/0); cap6 beats cap12 **12/0/1,
p=0.0005**; cap6?+R4 +0.040, **9/1/3**, p=0.0215 (R4's 1 loss = GX010137, a tolerance artifact ?
strict f1 & [?end] both IMPROVE there). Net guard **PASS**, merge_rate 0.651?0.161, **0 per-
video regressions**. Both new clips improve (0.448?0.685, 0.333?0.431). tIoU ladder 0.236?0.474.
14     - ? **Corroborated 2 ways:** sibling **#214 `lovo_resweep.py`** matches to 3dp + LOVO
```

cap=6 **13/13** on tIoU; my **5-lens adversarial workflow** `wf\_66c34d9d-057` ? **STANDS** (every L00 fold clears; worst-fold drop GX010128 still ?+0.202 p=0.0010 ? no single clip drives it). Reproduce: `scratch/r1\_evidence.py`.

15 - ? **#204 shipped clean:** rebased onto master (resolved a config/R5 conflict), 4 defaults flipped, 3 default-OFF tests + nightly docstrings updated, mandatory **docs/R1_MILESTONE_LAUNCH_REPORT.md** (both rulers + guard + caveats per [[launch-report-convention]]) + QUALITY_ITERATIONS "R1 LAUNCHED". Suite 855 green, cov 80.72%, ruff+mypy clean, 7/7 CI ? merged. **#216 re-baselines the nightly** (post-flip `default`==`milestone`; was self-flagging STALE ? re-snapshot n=13, both tolF1 0.359; data-only).

16 - **Caveats (Launch Report):** W1 cap = dominant lever / R4 = marginal +0.040; gate is the NET rate-based guard (strict=False BY DESIGN); GX010141 worst-behaved (future over-merge target); cap=4-vs-6 wobble = noise (LOVO re-fit to cap=4 is *worse*; tIoU firmly cap=6). **Optional follow-up:** add a pinned all-OFF reference config to the nightly to keep the pre-flip floor. ? clean worktrees verify-n13/r1-flip2. [[eval-dual-set-regression]] [[decision-rigor-norm]]

17

18 **Checkpoint:** 2026-06-18 (? **2 MORE BUG-FIX PRs SHIPPED & MERGED ? #212 + #213**;
found by adversarial bug-hunt after owner said "if there are any other fixes, send + merge their PRs"). After #209, ran a 2-pass adversarial hunt (by-area ? by-class + completeness critic, ~15 agents) over `origin/master`. Pass 1 = 0 findings (clean); pass 2 = 3 confirmed real defects (all owner_gated=false, I verified each against actual code, not trusting the agents):

19 - ? **#212 (`c7baf03`)? DB migration crash-safety:** v3/v4 used bare non-idempotent `ALTER TABLE ADD COLUMN` while v1/v2/v5 are `IF NOT EXISTS`. A DDL ALTER auto-commits independently of the `with conn:` scope, so an interrupted upgrade (after ALTER, before the end-of-block `PRAGMA user\_version` bump) leaves the column present + version stale ? next open re-runs the ALTER ? `OperationalError: duplicate column name` ? **`\_\_init\_\_` throws forever = permanently bricked DB**. Fix: `\_add\_column\_idempotent()` swallows ONLY 'duplicate column name'. Regression test proven to FAIL without the fix. Fully testable in CI (pure sqlite).

20 - ? **#213 (`bd33085`)? scripts unwrap segmenter Result:** `scripts/run\_phase3\_eval.py` (---segmenter` default==`motion`, the documented baseline) + `run_process_and_stitch.py` did `len()/truthiness on a raw `Success`/`Failure` dataclass (no `\_\_len\_\_`) ? `TypeError` on the default eval path; the `if not segments` guard was dead code. Other segmenters return bare lists (masked it). Fix: new generic `backend/results.unwrap\_or\_failure()` (DRY of backend/main.py's inline unwrap) used in both scripts + `tests/test\_results.py`. Verified vs the REAL motion segmenter. ? E2E motion pass needs GPU/cv2 (owner-side); fix correctness covered by unit test + real-segmenter Failure check + main.py parity.

21 - **DUE DILIGENCE each:** my full local gate (suite 848/851 pass, ruff, mypy 104, cov ~80.7%) + GitHub CI all 8 checks green + adversarial 4-lens review workflow per PR (both **verdict=merge, 0 must-fix**, independently re-verified). Addressed the reviews' should-fixes BEFORE merge (re-raise-branch test for #212; empty-Success test + no-overclaim docstring fix for #213). Worktrees/branches cleaned.

22 - ? **NOT auto-merged (left for owner/other sessions):** chip **PR #210** (sibling fusion_compare/fusion_audio loaders, same KeyError class) is STILL its spawned session's to merge ? I deliberately didn't touch it. Optional cleanup the hunt flagged (NOT bugs): fold the 4 hand-rolled Success/Failure unwraps (main.pyx2, generate_highlights.py) onto `unwrap\_or\_failure`; the litmus `>=6`/`=6` fragility ([[#209]] review). **NEW dev-env fact:** this container now HAS numpy/scipy/cv2/torch ? pure-logic eval AND segmenter-Failure paths run here (CLAUDE.md "lacks numpy/cv2/torch" is STALE; full GPU pipeline still can't run). [[working-agreement-merging]] [[decision-rigor-norm]]

23

24 **Checkpoint:** 2026-06-18 (? **ABLATION LITMUS MIXED-SCHEMA FIX ? MERGED #209 (`4bd9d2d`), issue #208 closed**). The pre-existing `test\_ablation.py` litmus `KeyError: 'shuttle'` (flagged 2026-06-17 as OUT-OF-SCOPE `task\_5878d224`) is FIXED + shipped to master. Root cause: `ablation.load\_videos()` read `c["shuttle"]`/`c["person"]` unconditionally, so once R3/R4 corpus growth added 5 clips persisted via the velocity-windowing/distill path (leaner rows, no cue dicts: Badminton_BXH_2, GX010128, GX030094, mahadevpura_1, mahadevpura_2) it crashed whenever the full n=11 corpus sat in `output/` (the 6 original fusion-schema files: adarsh_avi_singles, gbaaddy, kushagra_singles, largetest_doubles, mahadevpura_singles, testlarge_short). Fix (option a, most-robust): `\_is\_fusion\_schema()` predicate ? loader SKIPS leaner files (info-log, no crash); `\_golden\_present()` counts only fusion files so the skip-gate matches the loader. Litmus still reproduces Gate-A (?F1?0.074, 6/6, p?0.031 ? the 6 fusion files ARE the original n=6). +2 regression tests. **DUE DILIGENCE:** local full suite 848 pass/5 skip + ruff + mypy(104) + cov 80.66% (floor 70); **GitHub CI all 8 checks green**; **adversarial 4-lens review workflow ? verdict=merge, 0 must-fix** (independently re-verified the bug, the litmus reproduction, the 2-file scope). Owner reported issue #208 + opened PR #209 + squash-

merged on green. Worktree cleaned; local+remote branch deleted. ? **Sibling latent bug still open (`task\_8dd4efde`):** `fusion\_compare.load\_videos` + `fusion\_audio.load\_videos\_with\_audio` have the IDENTICAL unconditional `c["shuttle"]` access ? not caught by any test (synthetic tmp corpora only) but would crash the same CLIs on the mixed corpus. Separate PR (the review independently confirmed this scope call). Possible follow-up nit: litmus `>=6`-gate vs `==6`-assert would go red if a 7TH fusion clip is ever added (pre-existing; this fix improved the opposite drift). [[eval-dual-set-regression]] [[working-agreement-merging]]

25

26 **Checkpoint:** 2026-06-18 ~00:10 (? **REMAINING-3 4K REELS DONE ? ALL 6 BOXHILL-17 CLIPS NOW HAVE 4K** + cost-scaling doc + 10/10 visual QA ? all on G:). Owner asked for "just 4K highlights of the remaining 3, all local no Gemini." Delivered:

27 - ? **GX010138/139/140 4K reels** (the larger 9/10/12.8-min clips): **45/47/68 rallies**, 1.77/1.89/2.72 GB, 3840x2160, watermarked. Published to G: as `? - Highlights (4K).mp4`. Robust detached Scheduled Task `KhelSutraBoxhill4KRemaining` (keep-awake + self-restart; unregistered on completion).

28 - ? **DOGF00DED `--detect-from` end-to-end on fresh videos** (the merged #205 feature): `--video <4K original> --detect-from <1080p proxy>` ? detect on proxy, stitch 4K from original, reel named after `--video (NO `_1080p` naming bug this time). Log confirmed "detecting on proxy ? stitching reel from ?". Validated beyond the unit tests.

29 - ? **Visual QA workflow** (`wf\_2bd49aa1-361`, 10 vision agents): **10/10 delivered reels clean** ? correct resolution, watermark VISIBLY burned in, real rally content. (Adversarial verification of the shipped artifacts.)

30 - ? **`COST\_SCALING.md` (all 6 clips)** on G:. **Cost model (RTX 2070 Super Max-Q):** GPU detection (WASB) is the driver at **~3.4x real-time** (?3.4 GPU-min/footage-min, ~0.055 s/frame); NVENC proxy ~0.2x RT; 4K stitch (CPU libx264) ~16 s/rally; end-to-end ~5x clip length; peak VRAM ~5.7 GB (fits 8 GB). ? the per-video GPU-busy sampler OVER-counts (background desktop GPU during the CPU stitch counted as busy) ? use the WASB wall-clock as the clean GPU number. 6-clip totals: 52.2 min footage ? 176 GPU-min WASB + 68 CPU-min stitch, 250 rallies, ~11.6 GB of 4K reels.

31 - ? **GX010137 INGESTED as GOLDEN #13 (n=13) + SCORED (2026-06-18) ? best clip so far, big for the gap story.** golden 34 rallies (post-annotator-bug-fix labels). **LOCAL F1@0.5 0.82 (R 0.88!) vs GEMINI 0.89 (R 0.79):** on normal match play **local OUT-RECALLS Gemini** (finds more real rallies), but fires 14 FP vs Gemini's **0** (Gemini = perfect precision, misses 7). **? CONFIRMS opposite failure modes (local high-recall/low-precision, Gemini reverse) at n=2 ? PRECISION is local's #1 lever (Gate-A/colocation), fusion the durable win.** Local's gap to Gemini SHRINKS on normal play (hard clip ??0.16 ? normal ??0.07). Boundaries tight (local 0.39/0.44, tighter than Gemini). Folded into the gap-closure doc \$3 (PR #211, `f04f3f3`). ? **OWNER found + fixed ~3 rally-annotator BUGS** (GX010137 post-fix; **GX010141 may be PRE-fix ? re-export+re-score to confirm its less-flattering numbers**). Corpus aggregate n=13: tIoU F1@0.5 0.474, R2 tolF1 0.359. Scorer `scratch/boxhill17\_score\_golden.py` CLIPS now has both; add the next clip + re-run.

32 - ? **GOLDEN #14/#15 INGESTED + SCORED (2026-06-18, n=15): Video 12 Boxhill_Doubles (MULTICOURT, 43 golden) + Video 4 GX020094 (singles, 40 golden), both 29.97fps.** Pre-staged WASB trajectories (KhelSutraGoldenPrep task) made it instant. **PRECISION-LEVER SMOKING GUN ? multicourt Boxhill_Doubles: local fired 87 vs 43 golden, tolP 0.20 (?80% FALSE POSITIVES = background-court games!), tolR 0.40, F1@0.5 0.43, seg_ratio 2.02x.** Video 4 singles also over-fires (local 59 vs 40, tolP 0.31). Boundaries tight on both (|?|?~1s). **? 4 ground-truth clips deep, the story is identical: gap = PRECISION (over-fire, worst on multicourt) + some recall, NEVER boundaries.** Strongest case yet for Gate-A/colocation precision lever (gap-closure doc G2). NO Gemini on these yet (offer 3-way). Ingest via `scratch/ingest\_golden.py`; local in `output/golden\_prep.db`. Corpus n=15: tIoU F1@0.5 0.468, R2 tolF1 0.353. Prep task staging Video 1+9 trajectories (owner labeling 7+9 next; **Video 7=61min HELD pending scope confirm**).

33 - ? **LOVO THRESHOLD RE-SWEEP at n=13 ? thresholds HOLD, "retrain" buys nothing yet (MERGED #214 `55e4c89`, archived `docs/archives/quality-iterations/2026-06-18-lovo-resweep-n13/`).** Owner asked: now corpus is bigger, worth re-calibrating? Honest answer via leave-one-video-out: **tIoU picks cap=6 unanimously (13/13), ?(LOVO?fixed6) +0.000; R2 tolF1 cap=4 edges cap=6 by 0.005 but LOOCV re-fit doesn't generalize (??0.004=noise). R4 still significant (p=0.001/0.021); milestone?baseline (p<0.001).** ? corpus is in MEASURE (regression) territory not RETRAIN territory at this n; real retrain payoff = the learned fusion scorer, NOT these thresholds. **REUSABLE `scratch/lovo\_resweep.py`** (CPU-only, re-run after each label batch). Taught owner LOOCV-vs-k-fold (leave-1 while n small + folds cheap ? stratified k-fold at hundreds). Watch: cap-4-vs-6 tolF1 wobble. ? owner found+fixed ~3 rally-annotator bugs (GX010141 pre-fix ? re-export to confirm its numbers).

34 - ? **GX010141 INGESTED as GOLDEN #12 (n=12) + verified.** `scratch/ingest\_golden.py` (REUSABLE ? `--name/--golden-csv/--traj/--fps/--width/--tag`): copies

labels?`output/GX010141.rallies.csv` + traj?`Collect/Trajectories/GX010141\_june17\_traj.csv`, writes `output/GX010141\_golden\_features.json` (gts+candidates), appends `output/human\_lovo\_manifest.json` (n=11?12). ****Alignment OK**** (29 gts max 371.5s ? 385s trajectory). `rally\_seg\_eval` now scores it IDENTICALLY to the standalone scorer (tIoU@0.5 F1 0.510, R2 toIF1 0.431/P0.50/R0.38/|?start|0.78/|?end|0.68) ? clean integration. ****?**

`docs/GOLDEN\_VIDEOS.md` registry doc still trails reality** (shows n=9 + GX030094/BXH2 "in progress"; manifest is n=12 incl those 2 + GX010141) ? sync as a PR (batch with the next clips, or now). When GX010137/020140(+large) land: `ingest\_golden.py` each ? re-run `rally\_seg\_eval` (the milestone holds) ? one GOLDEN_VIDEOS.md sync PR.

35 - ? ****GOLDEN GX010141 LANDED + SCORED (2026-06-18, `F:\?\Badminton - Boxhill - 17 June\GX010141.rallies.csv`, 29 rallies) ? CORRECTS my earlier optimistic agreement read.**** Real accuracy (`scratch/boxhill17\_score\_golden.py` ? published `VS\_GOLDEN.md`): ****Gemini BEATS local on THIS clip**** ? tIoU@0.5 F1 Gemini ****0.67**** (P0.94/R0.52) vs local ****0.51**** (P0.59/R0.45); toIF1(??/1.5) Gemini 0.53 vs local 0.43. ****Local's over-production = FALSE POSITIVES, not missed-by-Gemini rallies**** (local 22 preds ? 13 tIoU-hits / 11 FP; Gemini 16 ? 15 hits / only 4 FP). ? ****GX010141 is the disagreement-OUTLIER I deliberately picked first (local's likely-WORST clip)**** ? do NOT generalize from n=1; the high-agreement clips (137/020140) next will likely favor local. Nuances: (a) BOTH miss ~half the 29 golden (recall ~45-52% ? detection is HARD on this continuous-drilling clip); (b) local & Gemini hit DIFFERENT golden rallies (only 3 overlap each other) ? ****complementary ? fusion could lift recall****; (c) ****boundaries TIGHT for both**** (local |?start|0.78/|?end|0.68s) ? R2/R3/R4 windowing holds; the gap is DETECTION (recall + local precision), NOT boundaries. Honest correction logged ([[working-agreement-attribution]]).

36 - ? ****OWNER labeling more (took a break): GX010137 + GX020140 next (the 2 with Gemini) + maybe 1 large-rally-count clip; then I INGEST the batch as golden + update `docs/GOLDEN\_VIDEOS.md`.**** Scorer is generic ? add each clip's `(stem, proxy, golden\_csv, has\_gemini)` to `CLIPS` in `boxhill17\_score\_golden.py` + re-run. video_ids/cached rallies: local GX010137=39/GX020140=29/GX010138=45/GX010139=47/GX010140=68; Gemini (first 3) 27/16/16. Labels apply 1:1 to 1080p+4K (same timeline). [[golden-set-vendoring]] [[eval-dual-set-regression]] [[eval-boundary-tolerance]]

37 - ****superseded by the above**** data-driven triage of the precision question (``peak_velocity`/`active_frame_count` cut`) is now moot once golden labels land. [[gotcha-long-runs-gpu-wsl]] [[paper-coauthor-aim]]

38

39 ****Checkpoint:**** 2026-06-17 ~20:55 (? ****WATERMARK BUG FIXED #205 (`8736e2d`) + 4K reels + GEMINI COMPARISON ? all published to G:****; Boxhill spawned-task follow-ups DONE). Owner reviewed the Boxhill reels and flagged 3 things; all addressed:

40 - ? ****#205 MERGED (`8736e2d`, owner-authorized merge):**** `fix(runner): honor stitching watermark/padding + add --detect-from`. ****ROOT CAUSE of missing watermarks:**** `tools/generate\_highlights.py` built `VideoCompiler(local\_work\_dir)` BARE ? `watermark=None` (off) + ctor `end\_padding=1.0` instead of `config.json stitching.{watermark, begin/end\_padding=0.5}`. Every other compile site threaded config; the runner didn't. Fix reads `cfg.get("stitching")` dict-compatibly + passes watermark+paddings. ****+ `--detect-from`****: detect on a cheaper proxy, stitch the reel from the full-res `--video` (the "default-to-original, flag-to-save-bandwidth" split owner described); 1:1 paired, reel named after `--video`. 3 new tests; ruff+mypy+844-pass+cov80.65% green; adversarial-review workflow (findings were intended-changes or pre-existing, none must-fix). ? ****LESSON:**** I'd told the owner the reels were "watermarked" WITHOUT verifying ? they weren't; owner caught it. Verify visual/output claims before asserting ([[working-agreement-attribution]]). Verified the fix by extracting a frame + eyeballing the mark.

41 - ? ****Reels REGENERATED + published**** (G:\?\Boxhill\2026-06-17 Local no-AI (improved windowing)): ****1080p WATERMARKED**** (re-stitched from cached rallies, overwrote un-watermarked) ****+ 4K WATERMARKED**** (`?` (4K).mp4, ~1.6?1.9GB each, stitched from the 4K originals; rallies 1:1 from DB, NO WASB re-run). Direct `VideoCompiler` re-stitch (`scratch/boxhill17\_restitch.py`) ? run_predict has no CSV short-circuit so re-running the runner risks re-inference; direct re-stitch is guaranteed-fast. NOTE end_pad 1.0?0.5 (config) makes reels ~0.5s×N shorter than the first cut.

42 - ? ****GEMINI COMPARISON**** (owner ask): ran Gemini on the 3 proxies (separate DB `pr-stitch/output/boxhill17\_gemini.db`, preserves local), stitched watermarked ****Gemini reels**** (`?` - Gemini.mp4) next to local + ****LOCAL_vs_GEMINI.md**** eval (`scratch/boxhill17\_eval.py`, reuses R2 `tolerance\_metrics`). ****KEY (agreement, NOT accuracy ? clips not golden):**** local ****90 rallies vs Gemini 59**** (local out-detects ALL 3); boundary agreement TIGHT (|?start|/|?end| <1s) on GX020140/GX010137 (the R2/R3/R4 windowing paying off); ****GX010141 = high-disagreement outlier**** (local 58% vs Gemini 19% coverage ? continuous drilling, ambiguous rally definition). Open Q: is local's +50% real rallies Gemini missed, or false positives? ? golden labels on these clips would settle it.

43 - ? Linger: worktrees `boxhill17` (WASB cache + `boxhill17.db` + reels), `pr-stitch` (gemini DB/reels + a copied `.env`) ? keep until owner done eyeballing; tasks
 KhelSutraBoxhill17/Gemini/4K unregistered. [[gotcha-long-runs-gpu-wsl]]

44

45 ****Checkpoint:**** 2026-06-17 ~18:40 (? ****R1 GUARD #201 + R5 BLOB-FALLBACK #203 SHIPPED****, master `f5f5e52`; ? ****R1 config-flip = DRAFT #204 awaiting OWNER PRODUCT-GO**** ? spawned R1 task complete). R-series: R1? R2? R3? R4? R5? all on master; the only remaining R1 step is the owner's shipped-behaviour flip.

46 - ? ****R1 net over-seg guard ? #201 merged**** (`8c68143`):
 `tolerance\_metrics.over\_seg\_guard` refines R2 §2.3.3's strict "no split/merge increase on ANY video" (which mis-fires on the structural mega?bounded change) to a ****NET guard on normalized RATES**** ? passes iff corpus-mean weighted ($w_{\text{merge}}=2w_{\text{split}}=1$) $\text{merge_rate} + \text{split_rate}$ cost doesn't rise AND no per-video merge_rate regresses. Wired into `rally\_seg\_eval.compare()`/`--vs` (+ self-describing header, `--vs-max-window-duration`). ****R1 EVIDENCE (n=11, R2 tolF1):****
 baseline?milestone(cap6+R4) ****? $+0.232$ [$+0.153, +0.304$], sign 10/0/1, p=0.002****; ****cap12?cap6 ? $+0.108$, sign 10/0, p=0.002**** (R3 confirmed via sign-test, was means-only); cap6?+R4 $+0.047$, 8/0, p=0.008. mean merge_rate 0.694?0.150 (all 11 ?); ****net guard PASS, strict BLOCK**** (the artifact). `scratch/r1\_evidence.py`.

47 - ? ****R5 blob-fallback ? #203 merged**** (`f5f5e52`, default-OFF):
 `tracknet\_runner.\_blob\_fallback\_chop` ? AFTER the gap-split + R4 trim, midpoint-chop any group still > ****blob_factor(4x)**** the cap (the gap-less blob), ****log + flag `forced\_cut`****. The factor gate keeps it surgical (vs `hard\_cap` which chops before R4 + over-segments normals). ****mahadevpura_1 blob tolF1 0.000?0.238**** (2 windows max 65.8s ? 32 max 4.1s); do-no-harm elsewhere at factor 4 (stratum tool, agg $+0.030$ n.s.). Wired config/preset/served/CLI `--blob-fallback`.

48 - ? ****R1 config-flip = DRAFT PR #204**** (`feat/r1-milestone-config-flip`, NOT merged ? shipped-behaviour change): flips IndexingConfig defaults `max\_window\_duration` 0?6, `window\_inactivity\_end` 0?1.5, `inactivity\_rally\_thresh` 0.08?0.20, `inactivity\_min\_density` 0.34?0.30. Config-only; ****merging entails**** 3 default-OFF test updates + the ****nightly coupling**** (`nightly\_regression.tracked\_configs` "default"=SERVED ? post-flip default==milestone, all-OFF drift baseline collapses ? decide: drop redundant milestone cfg, or re-pin an explicit all-off control). Left for owner; I finish the test/nightly updates once the direction is chosen.

49 - ? ****DISPATCH (owner): R1 PRODUCT-GO**** ? flip the milestone default ON (review/merge draft #204 after deciding the nightly-control question). Evidence clears the net-guard bar ($+0.232$, 10/0, p=0.002). Also still open: ****IAA study**** (2nd labeler re-labels 2?3 golden clips ? fix R2 ?). [[eval-dual-set-regression]] [[eval-boundary-tolerance]] [[decision-rigor-norm]]

50 - ? Linger worktrees to clean later: r1-guard, r5-blob, r1-flip, nightly-reg (sibling) + the older ones listed below.

51

52 ****Checkpoint:**** 2026-06-17 ~17:50 (? ****NIGHTLY RALLY-QUALITY REGRESSION SHIPPED ? PR #202 MERGED****, master `2eaccd5` ? spawned task #3). The owner's "process videos every night as nightly tests" ask, as a CPU ****tripwire**** (NOT a re-training run):
 `scripts/nightly\_regression.py` re-scores the golden corpus (n=11) under ****DEFAULT**** (SERVED, knobs OFF) + ****MILESTONE**** (R3 cap=6 + R4 inactivity-trim) via `rally\_seg\_eval.evaluate` (cached trajectories ? seconds, no GPU/Gemini/WASB) and diffs headline metrics + the per-video over-seg guard vs frozen `eval\_baselines/nightly\_baseline.json`. Frozen n=11: DEFAULT tolF1 0.091/f1@0.5 0.194; MILESTONE tolF1 0.323/f1@0.5 0.440 (matches QUALITY_ITERATIONS R3/R4 ? cross-checks the harness). Exit 1=regression, 4=stale(corpus/config changed?`--update-baseline`); writes `output/nightly\_regression/<date>.{md,json}`. ****DEFAULT-ADDITIVE ? does NOT change the promotion gate.**** 27 offline unit tests; verified full suite+ruff+mypy green (cov 80.69%), e2e on real n=11 (both configs PASS) + tripwire fires on simulated regressions.

53 - ****Scheduled task registered:**** `KhelSutraNightlyRegression` (daily ****03:00****, user-level, StartWhenAvailable) ? `scripts/nightly\_regression.ps1` wrapper, pointed at the MAIN checkout. ****Inline-guarded ? currently no-ops (SKIPPED)**** because the main checkout is on the sibling branch (95cd154), not master; it SELF-ACTIVATES once the main checkout is on master** (per the clean-slate convention). Re-point/remove via `scripts/register\_nightly\_task.ps1`. A cloud agent can't run it (cached trajectories live outside the repo).

54 - ? ****DISPATCH (owner):**** when you next put the main checkout on master, the nightly goes live automatically (no action needed); to run it NOW: `Start-ScheduledTask KhelSutraNightlyRegression` from a master checkout, or `python scripts/nightly\_regression.py`. Re-snapshot the baseline (`--update-baseline`) after R1 flips the milestone ON or when the golden corpus grows.

55 - ? ****Pre-existing, OUT-OF-SCOPE (spawned task `task\_5878d224`):**** with the full n=11 corpus in `output/`, 2 `test\_ablation.py` litmus tests fail LOCALLY (`KeyError: 'shuttle'`) ?

the 5 newer real clips' `\_golden\_features.json` carry a leaner velocity-windowing schema than the n=6-era litmus expects. CI unaffected (empty `output/?skipped`); the nightly path is immune (reads only `gts+trajectory`). [[eval-dual-set-regression]] [[working-agreement-merging]]

56

57 ****Checkpoint:**** 2026-06-17 ~18:15 (? ****BOXHILL-17 3-VIDEO HIGHLIGHTS ? DONE + PUBLISHED** to G:**** ? spawned task #2 COMPLETE**). LOCAL no-AI reels for the ****3 shortest**** Boxhill-17-June clips: ****GX010141 (6:25), GX020140 (6:26), GX010137 (7:31)**** (the other 3 ? GX010138/139/140 ? are 8-11.5GB, skipped to keep GPU time reasonable). Improved milestone windowing in worktree ``.claude/worktrees/boxhill17`` config.json (``max_window_duration=6.0``, ``window_inactivity_end=1.5``, ``inactivity_rally_thresh=0.20``, ``inactivity_min_density=0.30``, ``wasb.timeout_sec=5400``; verified served preset reads them, ``window_hard_cap`` stays False ? soft cap-6 = R3 op-point). Branch ``run/boxhill17-highlights@`34b9caa`` (origin/master, R4). Flow per video: NVENC 4K?1080p proxy (``output/boxhill17_proxies``) ? runner ``wasb_hybrid --skip-ai-handoff --wasb-stream --db output/boxhill17.db`` ? reel to ``output/boxhill17_reels`` ? ****published to** ``G:\My Drive\Sharing\Badminton\Boxhill\2026-06-17 Local no-AI (improved windowing)\``****** (additive). GX010141: proxy ~80s (~4.8x realtime), WSL env OK (torch+CUDA), WASB inference running @17:02. ROBUST LAUNCH: detached ****Scheduled Task `KheISutraBoxhill17`**** (keep-awake ``SetThreadExecutionState`` + self-restart, mirrors ``extract_val_until_done.ps1``); wrapper ``scratch/boxhill17_highlights.ps1``; ****5-min heartbeat Monitor `bop8jl2j4`****. COST: per-video ``_COST_REPORT.md`` (wall-clock/phase + GPU-busy-min via 10s nvidia-smi sampling + peak VRAM) in ``output/boxhill17_cost``. To resume/observe: ``tail `scratch/boxhill17_wrapper.log``; ``Get-ScheduledTask KheISutraBoxhill17``. Reels are 1080p (proxy-based); 4K stitch-from-original is a cheap follow-up (rallies already in DB). ? ****RESULT:**** all 3 published ? ****GX010141 22 rallies/4:18, GX020140 29/3:42, GX010137 39/4:07**** (90 total; h264 1080p watermarked, 257/222/247s; 55-67% of each clip kept). ****COST (n=3, ~20.4min 4K footage, RTX 2070 Super Max-Q):**** GPU-busy ****67.2 min**** total, WASB ****65.1 min**** (~3.2x clip duration ? ~1 GPU-min per ~18s footage), proxy 4.2 min, peak VRAM ****5966 MB****. Per-video ``_COST_REPORT.md`` + ``README.md`` on G:. ? ****LESSON** (filename bug): ****the `\_1080p` proxy suffix made the runner's reel-stem (`GX010141\_1080p`) ? the wrapper's stem (`GX010141`) ? the wrapper's INLINE cost-report/publish/skip looked for the wrong filename (cost reports blank, no auto-publish, would re-loop). WASB output was fine; fixed POST-HOC via `scratch/boxhill17\_finalize.ps1` (regen cost from run.log + gpu.csv.bak snapshots + atomic publish). ****NEXT TIME: name proxies EXACTLY like the original**** (e.g. ``GX010141.MP4`` in a proxies dir, as the kushagra June-12 run did) so runner-stem==wrapper-stem. Scheduled Task ``KheISutraBoxhill17`` unregistered; heartbeat self-terminated; worktree config.json edit left uncommitted (isolated, intentional). [[gotcha-long-runs-gpu-wsl]] [[feedback-tail-long-job-logs]]**

58

59 ****Checkpoint:**** 2026-06-17 ~16:00 (? ****R4 SHIPPED #200 `34b9caa`****; ? ****CONTEXT-PAUSE + SPUN OFF** the rest****** per owner's explicit "ensure context window constraints, pause + spin off others"). R-series so far: ****R2? (#199) R3?(finding, cap6) R4? (#200)**** ? all on master. This session got very long ? paused; remaining work moved to fresh-context spawned tasks.

60 - ****NEW owner asks (2026-06-17, excited):**** (a) ****full highlights for 3 videos**** from ``F:\GoPro Hero Black 12\KheISutra\Badminton - Boxhill - 17 June`` (6 clips: GX010137/138/139/140/141, GX020140) ? AFTER all R-iterations, using the improved milestone windowing (cap=6 + R4). (b) ****Computational-cost measurement**** (CPU-min + GPU-min per video) ? he's still mulling ("let me process it more"); instrument per-phase wall-clock + GPU-busy in the highlights run. (c) ****An "educated quality LAUNCH" = flip the milestone defaults ON (this IS R1; he's now leaning yes ? still his product-go).** (d) ****Nightly regression set**** ? a scheduled task that re-scores the golden corpus nightly to catch regressions (his dual-eval-set discipline; note: cached-trajectory RE-SCORE is CPU+fast, no need to re-run WASB nightly).

61 - ****SPUN OFF** (3 fresh-context tasks, each reads this file + the docs): ******(1) R1 launch evidence + net-over-seg guard + R5 blob-splitter; (2) Boxhill-17-June 3-video highlights + cost instrumentation; (3) nightly golden-regression scheduled task. Each branches from ``origin/master`` (now ``34b9caa``).

62 - ****? DISPATCH (owner):**** (1) ? ****R1 LAUNCH APPROVED**** ? owner is happy to merge the launch PR GIVEN the documented impact table (below), which is now provided. Owner wants it (a) ****documented separately for posterity**** (last quality-bit flip ? Gate-A #146 ? noted a regression, hence the caution) and (b) ****merged by Claude**** (the R1 spawned task ``task_eac3f171``: ? confirm the sign-test + net-over-seg guard; ? write a ****LAUNCH REPORT**** ? owner's new **STANDING** convention for EVERY default config change, see [[launch-report-convention]] ? with the total-quality-impact table below + the Gate-A-lesson rationale, e.g. ``docs/launches/2026-06-17-rally-windowing.md`` + a **QUALITY_ITERATIONS** pointer; ? make the config-flip PR by editing the tracked ``config.json`` indexing block ? NOT the `IndexingConfig` dataclass defaults, which stay OFF so the eval substrate + default-OFF tests are unchanged ? and ****MERGE** it on green CI******. ? The original chip prompt said "leave the PR for owner go / do NOT flip";

THIS MEMORY SUPERSEDES that ? the product-go is GRANTED conditioned on the impact table, so create + merge it). (2) ****IAA study**** ? owner WILL find a 2nd annotator (endorsed it as good practice) ? fix R2 ?.

63 - ? ****LAUNCH CONFIG** = W1 cap=6 + R4 (inactivity_end=1.5, rally_thresh=0.20, min_density=0.30); NO hardcap (it over-splits ? lowers tolF1 0.323?0.288), NO W2 (opt-in/retained). ****Blob clip (mahadevpura_1)** handled by R5, not by adding hardcap to the launch. ****TOTAL LAUNCH IMPACT** (golden n=11, baseline?launch): ****tIoU-F1@0.5 **0.194?0.440 (+0.246)**** . R2 tolF1 ****0.091?0.323 (+0.232)**** . mean |?end| ****52.1s?3.8s**** . merge-count ****4.9?1.5**** (over-merge crushed); split 0?5.1 (the honest over-seg tradeoff, net tolF1 hugely +). Far more rigorous than the Gate-A flip (measured on 2 rulers + n=11 + R4's 8/0 sign-test p=0.008).

64 - ? Lingering worktrees to clean later: w2-safeguard, hardchop, docs-*, r4-endrule, gpu-validation, + older ones. Heartbeat monitors/Scheduled Tasks (KhelSutraValExtract, KhelSutraNewClipsExtract) are done ? can be unregistered. [[eval-dual-set-regression]] [[eval-boundary-tolerance]]

65

66 ****Checkpoint:**** 2026-06-17 ~15:00 (? ****MULTI-DAY AUTONOMY MANDATE****: owner greenlit executing ALL R* I deem possible while he's busy ~2 days adding 10-12 golden labels + talking to first customers. "Dispatch via app/here when you need something." Trusts repo sanctity to me. ? keep the discipline: measured, default-OFF, PRs clear the bar, dispatch when blocked.)

67 - ? ****R2 SHIPPED ? #199 merged**** (`7e8f88f`): `backend/eval/tolerance\_metrics.py` (strict-1:1 Hungarian tolerance-match, scipy-or-greedy; unconditional best-overlap boundary-error split start/end; over-seg guard; ?-sweep) + wired into `rally\_seg\_eval` (R2 block in score_intervals/aggregate/format_report/compare ?tolF1 + `--tau-start/--tau-end`). Default-ADDITIVE (tIoU still the gate pending the ablation experiment). Provisional ?_start=2.0/?_end=1.5. n=9 R2 aggregate: tolF1 0.197, |?start| 1.25s vs |?end| 2.19s (end=gap confirmed). R2 DESIGN doc #198 merged (owner locked the 3 decisions: asymmetric ?; IAA-after-code; augment?ablation-gate?deprecate-with-docs).

68 - ? ****R3 MEASURED (cap-sweep, n=9):**** ****12s cap was too coarse ? cap?6 is best**** (tolF1 cap6 0.273 > cap12 0.171). The |?end|-vs-split tension IS the R4 spec: plain W1 = loose ends (|?end|-7s, few splits); hard-cap = tight ends (~1s?Gemini, heavy over-split). ****R4 target:** hard-cap's tight ends WITHOUT the over-split** = end at the real inactivity point. (cap=6 is a means-lead; confirm via paired sign-test on n=11 before locking ? feeds R1's config value 12?6.)

69 - ? ****R1 BLOCKER (measured, key insight):**** baseline?W1+hardcap is a SIGNIFICANT tolF1 win (?+0.095, sign 8/1, p=0.039) AND kills merges everywhere (?0) ? BUT trips the strict "no split regression on ANY video" guard (split 0?4-14 on all 9: it trades merges for over-splits). ? R1 unblock options: (1) do R3+R4 first (reduce over-prod), (2) ****refine guard to NET over-seg**** (the honest guard; I'll implement this for R1), (3) ship as floor-raiser. Recommended owner: (2)+ship. R1 final flip = owner product-go (dispatch).

70 - ? ****R3 + R4 MEASURED & VALIDATED (n=11, under R2):**** R3 cap ****12?6**** (tolF1 0.171?0.276). ****R4 rally-END inactivity trim**** (`\_trim\_inactive\_tail` in tracknet_runner, default-OFF: trims post-rally slow-drift tail ? frames stay "active" while shuttle drifts >velocity_thresh but <inactivity_rally_thresh; trim to last frame whose trailing window is ? min_density` fast). ****Sweet spot T=1.5s, rally_thresh=0.20, min_density=0.3:**** cap6?+R4 ****?tolF1 +0.047 [+0.022,+0.072], sign 8/0, p=0.008****, |?end| 5.71?3.81s, splits NET-DOWN (only mp2 +1; GX010128 4?2, BXH2 5?4). Beats hard-cap's mechanism (hardcap got |?end| 1.07 but split 19.5 over-prod). Cumulative tolF1: baseline 0.096 ? cap12 0.171 ? cap6 0.276 ? +R4 0.323. R4 code in worktree `feat/r4-inactivity-end` (NOT yet wired to preset/config/CLI ? that's the in-progress PR).

71 - ?? ****NEXT (executing autonomously):**** finish ****R4 PR**** (wire `window\_inactivity\_end`/`inactivity\_rally\_thresh`/`inactivity\_min\_density` through WindowingPreset+IndexingConfig+trajectory_hybrid+rally_seg_eval CLI + tests + QUALITY_ITERATIONS) ? re-check ****R1**** (cap=6 + R4 + net-over-seg guard) ? ****R5**** blob-splitter. ****DISPATCH LIST for owner:**** (1) R1 product-go when it clears; (2) ****IAA study**** (2nd labeler re-labels 2-3 golden clips ? fix ? ? I can't generate IAA myself); (3) FYI net-over-seg guard tweak (revert=1 line).

72 - ? ****n=11 dessert in flight:**** BXH2 (Badminton BXH 2, 44 golden) WASB ~done via Scheduled Task `KhelSutraNewClipsExtract` (heartbeat `bvoc96z90`); GX030094 (41 golden, 30fps) already at-par'd (3rd venue confirms detection~80%/boundary~51%/Gemini-under-detects pattern + Gemini missed 14). When BXH2 lands ? ingest both as golden #10/#11 (n=11) + at-par + update GOLDEN_VIDEOS.md (#197). Gemini for both already in `output/sports\_indexer.db` (job bbofujewg done). [[eval-boundary-tolerance]] [[eval-dual-set-regression]] [[decision-rigor-norm]]

73

74 ****Checkpoint:**** 2026-06-17 ~14:00 (R2 DAY: golden corpus ****n=9****; golden-tracker + R2 design docs shipped; 2 new clips extracting). Owner back, wants R2 today + focus on LOCAL on-par-with-Gemini. Master now `c483346`.

75 - ****3 new `[Done]` badminton videos labeled**** (in `~/Golden Label Videos Request - June 15/`): Video 14=****mahadevpura-1**** (10 rallies, the blob), Video 6=****GX030094**** (41 rallies, ****29.97fps****), Video 10=****Badminton BXH 2**** (44 rallies). (Video 13=**GX010128** + Video 3=**mahadevpura-2** = already-ingested dupes.)

76 - ? ****mahadevpura-1** ingested as golden #9** + at-par run: it's local's WORST clip ? baseline/W1/W1+W2 score ****F1 0.000**** (1-2 mega-windows match nothing); ****hard-cap** rescues to 0.294/0.588** (F1@0.5/@.25) vs Gemini 0.842. Clearest proof hard-cap is non-optional for the milestone.

77 - ? ****GX030094 + Badminton BXH 2: WASB extracting**** via robust Scheduled Task `KhelSutraNewClipsExtract` (`scratch/extract\_newclips\_until\_done.ps1`, keep-awake + self-restart; heartbeat monitor `bvoc96z90`). GX030094 video_id `27f1c707bbfb`. ~1hr. When done ? Gemini + at-par + ingest as #10/#11 + update GOLDEN_VIDEOS.md. (Task registered WITHOUT -RunLevel Highest ? Highest needs admin this session lacks.)

78 - ? ****#197 MERGED**** ? `docs/GOLDEN\_VIDEOS.md` golden-labeled video registry (owner will isolate them). ? ****#198 OPEN for OWNER REVIEW**** ? `docs/R2\_EVAL\_METRIC\_DESIGN.md` (do NOT auto-merge; owner wants a deeper look).

79 - ? ****R2 PREVIEW FINDING (reshapes the design):**** a symmetric $\pm 2s$ tolerance under strict Hungarian 1:1 is NOT more lenient than tIoU ? it makes local look FURTHER behind (correctly charges the 1.5x over-production + loose ends tIoU's overlap-matching absorbed; GX010128 local/Gemini ratio 0.56@tIoU0.5 ? 0.41@?2.0). Even Gemini scores low at tight ? (mp2 tolF1@1.5=0.20) ? ****golden END convention carries >1.5s noise ? ?_end MUST be IAA-calibrated**** (2nd-labeler study). R2 design = ****asymmetric ? (?_start?2 forgives service window, ?_end?1.5 exposes the end-deficit) + Hungarian 1:1 + decomposition (P/R/F1@? + start/end boundary-error split + over-seg guard + ?-sweep)****; tIoU?secondary. `tolF1@~3.0?tIoU@0.5`.

80 - ?? NEXT: owner reviews #198 ? implement R2 (`backend/eval/tolerance\_metrics.py`, default-additive) ? R4 rally-end rule. Finish ingesting the 2 new clips. More golden coming (TT+pickleball later; badminton focus). [[eval-boundary-tolerance]] [[eval-dual-set-regression]]

81

82 ****Checkpoint:**** 2026-06-16 ~22:30 (RALLY-QUALITY: ? GX010128 GROUND TRUTH IN ? END-dominance CONFIRMED at n=2; ****golden corpus n=8****). Owner hand-labeled GX010128 (52 rallies) ? verified byte-identical to the windowed proxy (45,298 frames, perfect alignment), added as golden #8 (`scratch/at\_par.py` is the reusable analyzer). ****KEY RESULTS** (2nd ground-truth venue, vs 52 golden):**

83 - ? ****END-dominance REPLICATES** (now n=2, no longer a fluke):** local |?start| med ****0.78s**** (?/better than Gemini 0.98s ? START at parity) but |?end| med ****2.12s vs Gemini 0.76s**** (the whole gap). Same as mp2 (start 1.74?1.46, end 3.12 vs 1.19). ? ****R4 sustained-inactivity rally-END rule is the LOCKED next lever; NO serve/START classifier (R8 ? START solved).****

84 - ? ****Local OUT-RECALLS Gemini here:**** local W1+W2 overlaps ****all 52**** golden rallies; ****Gemini MISSED 13**** (matched 39/52, R 0.71). Local's gap is ****precision + loose ends**** (75 windows, P 0.39, F1@0.5 0.457), NOT detection. ? corrects the report's "34 GX010128 false-positives" (?13 were real rallies Gemini missed; ground truth disentangled it).

85 - ****Overall F1@0.5:** Gemini 0.813 (P 0.95!) vs local-best W1+W2 0.457**; F1@0.25 Gemini 0.857 vs local 0.756 (~88%, matches mp2's ~86% detection). Gemini wins the headline via tight precision; local wins recall.

86 - ? ****Best local config is CLIP-DEPENDENT:**** GX010128 ? ****W1+W2 best**** (0.457), hardcap OVER-splits this gap-rich clip (0.421, 7 splits). mp2 ? hardcap best (continuous-blob). Confirms "no single global config; hardcap is content-adaptive" ? and that ****W2's value is also clip-dependent**** (helped GX010128 +0.011, hurt recall on mahadevpura-0). Milestone still owner-gated, re-run at n?16.

87 - ?? NEXT: implement ****R2 eval metric**** ($\pm 2s$ tolerance-F1 + Hungarian 1:1 + over-seg guard + boundary-error split) ? it'd credit local's recall + START accuracy that tIoU 0.5 hides; then ****R4 rally-END rule****. Fold REAL_FOOTAGE_VALIDATION_REPORT.md into RALLY_QUALITY_RESEARCH. Weekend +golden (toward ~16) ? lock the milestone via sign-test sweep. [[eval-boundary-tolerance]] [[decision-rigor-norm]]

88

89 ****Checkpoint:**** 2026-06-16 ~20:50 (RALLY-QUALITY: ? FULL-6 REAL-FOOTAGE VALIDATION DONE + adversarially-verified report). All 3 missing trajectories extracted (robust Scheduled Task held ? NO second silent death); full-6 ran; multi-lens Workflow `wf\_736be611-fb5` (17 agents, killed 6/12 claims) ? `output/REAL\_FOOTAGE\_VALIDATION\_REPORT.md`. ****Adversarial pass CORRECTED my framings:****

90 - ? ****Milestone recommendation is W1 cap=12 + W1.5 hard-cap (NO W2)**** ? NOT "W1+W2" as I'd said. W2 (net-crossings) costs ****recall (6 net-new gemini-only misses, ~all on mahadevpura-0 62?43) for NO ground-truth-backed gain**** (golden +0.019 n.s.) ? ****keep W2 default-OFF + retained**** ([[weak-signals-retained]]), do NOT promote. Hard-cap is do-no-harm (fires only when

W1 finds no gap ? no-op on the 4 gap-segmented clips) + rescues the blob + best on the GT clip.
Even W1+hardcap ship is **gated on a re-run at ~16 clips** (hard-cap was golden-neutral n=6,
best n=7 ? fragile).

14. user (2026-06-18T07:43:51.089Z)

```
1      """WASB shuttle-trajectory runner (WSL2 subprocess adapter).
2
3      Windows-side counterpart to backend/detectors/wasb_infer.py. It:
4          1. stages the video into the WSL filesystem (fast local reads, not /mnt),
5          2. copies wasb_infer.py into the WASB repo's src/ dir,
6          3. runs it in the WSL `wasb` conda env -> a Frame,Visibility,X,Y trajectory CSV,
7          4. reuses the (model-agnostic) trajectory->action-window logic from tracknet_runner.
8
9      So the only WASB-specific bits live here; the windowing + CSV parsing are shared
10     with the TrackNet path. Verified inference core: wasb_infer.py (see its docstring).
11     """
12     import os
13     import shlex
14     import logging
15     import subprocess
16     from dataclasses import dataclass
17     from typing import Dict, Any, Optional, Tuple
18
19     # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical.
20     # parse_trajectory_csv / trajectory_to_action_windows are @staticmethods on
21     TrackNetRunner.
22     from backend.pipeline.detectors.tracknet_runner import to_wsl_mnt_path, TrackNetRunner
23     from backend.pipeline.detectors.base import DetectorRunner, WslCommandMixin,
24     wsl_tilde_quote
25
26     logger = logging.getLogger(__name__)
27
28     # Windows path to the inference wrapper we stage into WSL.
29     _INFER_WIN = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
30
31     @dataclass
32     class WasbConfig:
33         """Resolved settings for a WASB WSL run (built from config.json ->
34         indexing.wasb)."""
35         distro: str = "Ubuntu"
36         repo_dir: str = "~/models/WASB-SBDT"
37         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
38         conda_env: str = "wasb"
39         weights_path: str = "~/models/WASB-
40         SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
41         sport: str = "badminton"
42         wsl_stage_dir: str = "~/clips"
43         timeout_sec: int = 1800 # 30 min; a hung GPU/conda call is killed (0 = no timeout)
44         keep_frames: bool = False # retain staged video + frame cache after success (debug
45         only)
46         min_free_gb: float = 0.0 # warn if WSL free space below this before a run (0 =
47         off)
48
49         # FAST PATH (default OFF until verified): decode the staged video on the fly inside
50         # WSL and feed frames straight to the detector, skipping the slow per-frame PNG
51         # extraction that dominates a long clip. Output is bit-identical to the PNG path
52         # (PNG is lossless), but this stays opt-in until the owner's side-by-side confirms
53         it.
54         stream_video: bool = False
55
56     @classmethod
57     def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "WasbConfig":
58         w = (idx_cfg or {}).get("wasb", {}) or {}
59         return cls(
```

```

53         distro=w.get("wsl_distro", "Ubuntu"),
54         repo_dir=w.get("repo_dir", "~/models/WASB-SBDT"),
55         conda_sh=w.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
56         conda_env=w.get("conda_env", "wasb"),
57         weights_path=w.get("weights_path",
58                             "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"),
59         sport=w.get("sport", "badminton"),
60         wsl_stage_dir=w.get("wsl_stage_dir", "~/clips"),
61         timeout_sec=int(w.get("timeout_sec", 1800)),
62         keep_frames=bool(w.get("keep_frames", False)),
63         min_free_gb=float(w.get("min_free_gb", 0.0)),
64         stream_video=bool(w.get("stream_video", False)),
65     )
66
67
68 class WasbRunner(WslCommandMixin, DetectorRunner):
69     def __init__(self, cfg: WasbConfig):
70         self.cfg = cfg
71
72     def healthcheck(self) -> Tuple[bool, str]:
73         """Verify the WSL `wasb` env imports torch and sees a GPU. Returns (ok, msg)."""
74         cmd = (f"conda activate {self.cfg.conda_env} && "
75               f"python -c \"import torch; print('torch', torch.__version__, "
76               f"'cuda', torch.cuda.is_available())\"")
77         try:
78             res = self._wsl_bash(cmd)
79         except FileNotFoundError:
80             return False, "`wsl` not found ? Windows + WSL2 required."
81         except subprocess.TimeoutExpired:
82             return False, "WSL healthcheck timed out."
83         out = (res.stdout or "").strip()
84         if res.returncode != 0:
85             return False, f"WSL `wasb` env check failed: {(res.stderr or out).strip()}"
86         return True, out
87
88     def _stage_video(self, video_win_path: str, dest_name: str) -> Optional[str]:
89         src_mnt = to_wsl_mnt_path(video_win_path)
90         reason = self.wsl_source_unreadable_reason(src_mnt)

```

15. assistant (2026-06-18T07:44:00.992Z)

Let me ground the de-WSL crux: read the detector base (WSL mixin), the actual inference core, the local endpoint, and locate the packaging item + owned-model tie-in.

16. user (2026-06-18T07:44:01.956Z)

```

1     """Abstract base for detector runners (TrackNet, WASB, future native, etc.).
2
3     This completes the Detector Runner Abstraction (low-regret item 6 continuation / item 1
4     of the post-observability re-prioritization) to de-risk the WSL/conda dependency for
5     the high-accuracy shuttle trajectory substrate used by tracknet_hybrid / wasb_hybrid.
6
7     The ABC lets us:
8     - Keep the current WSL adapters working unchanged for Windows users.
9     - Swap in (or A/B test) a Linux-native implementation later with zero changes to
10    the hybrid segmenters or the model-agnostic windowing brain.
11    - Provide a single place to evolve the healthcheck / error contract and thread
12    detector readiness into per-video reports.
13
14    Contract (deliberately matches what the concrete runners + hybrids + tests already did):
15    - run_predict(video_win_path: str, output_win_dir: str) -> Optional[str]
16      Run detector, return the Windows-side path to the produced trajectory CSV
17      (e.g. ".../clip_predict.csv" or ".../clip_wasb.csv"), or None on any failure.
18      The caller (hybrid) is responsible for output dir; impls may stage the video

```

```

19     into WSL fs for perf.
20
21 - healthcheck() -> tuple[bool, str]
22     Return (ok, message). Never raises. "ok" decides whether to proceed to run_predict.
23     Message is logged (and can be surfaced to reports or UX in future).
24     Intended to be cheap (env existence, import, GPU visibility, weights present).
25
26 Existing behavior is 100% preserved. The only additions are the inheritance and
27 the explicit common type for future implementers.
28
29 How to add a new runner (for docs / future contributors):
30 1. Subclass DetectorRunner in a new or existing module under pipeline/detectors/.
31 2. Implement run_predict and healthcheck (return (bool, str) tuple).
32 3. If it produces the same CSV schema (Frame,Visibility,X,Y), reuse or re-export
33     TrackNetRunner.parse_trajectory_csv and .trajectory_to_action_windows (they are
34     deliberately static and model-agnostic). If the new detector needs different
35     windowing, it can implement its own or we generalize later.
36 4. Lazy-import heavy native deps inside the methods (so importing the segmenter
37     registry never pulls torch/tf on a machine that doesn't have them).
38 5. Update the two hybrid segmenters (or introduce a detector= choice) to accept
39     DetectorRunner instances (they already do the right thing via duck-typing + the
40     type hints we add here).
41 6. Add unit tests that exercise the contract (isinstance, healthcheck shape, run
42     returns str|None) using mocks for the external bits ? see test_tracknet_runner.py.
43 7. No change to eval/calibration code paths (they consume the CSV + the static
44     windowing helpers directly).
45
46 Optional Linux-native path (the main de-risk goal):
47 A LinuxNativeRunner (or ONNXRunner) would:
48 - Skip all wsl / _stage / conda activate.
49 - Load weights locally (torch or onnx) or call a native binary.
50 - Write the exact same CSV format so trajectory_to_action_windows works verbatim.
51 - healthcheck can just check "weights file exists and torch importable + CUDA?".
52 This removes the WSL requirement for Linux users / future SaaS workers while
53 keeping the exact same candidate quality characteristics (or better, once native
54 weights are available).
55
56 Error surfacing into reports:
57 The healthcheck message is already logged by the hybrids on failure. The reporting
58 harvest sees the outcome as "0 candidates" + detector name. When we wire richer
59 segmenter Results (or pass detector_health into emit_indexer_report context), the
60 (ok, msg) can be recorded explicitly under segmentation.details or a new detector
61 stage. The ABC makes that future change local to one place.
62
63 See also:
64 - backend/pipeline/segmenters/{tracknet_hybrid,wasb_hybrid}.py (the two callers)
65 - tests/test_tracknet_runner.py (contract + pure logic tests; hybrids mock the runner)
66 - docs/TRACKNET_WSL_SETUP.md and CODE_MAP.md (2.2 Detectors section)
67 ""
68
69 import logging
70 import shlex
71 import subprocess
72 from abc import ABC, abstractmethod
73 from typing import Any, Callable, List, Optional, Tuple
74
75 logger = logging.getLogger(__name__)
76
77
78 def wsl_tilde_quote(path: str) -> str:
79     """shlex.quote that still lets bash expand a leading ``~/``.
80
81     ``shlex.quote("~/clips/x.mp4")`` single-quotes the whole path, so bash
82     treats ``~/`` literally and ``cp``/``mkdir`` fail with "No such file or
83     directory" ? this broke live WSL staging for the default ``~/clips``

```

```

84     stage_dir (found by the 2026-06-11 wasb_hybrid smoke run; the BACKLOG
85     "shlex-quote breaks ~ expansion" item). Quoting only the part after
86     ``~/`` keeps expansion AND protects spaces/specials in the remainder.
87     """
88     if path == "~":
89         return path
90     if path.startswith("~/"):
91         return "~/ " + shlex.quote(path[2:])
92     return shlex.quote(path)
93
94
95     class WslCommandMixin:
96         """Shared ``wsl -d <distro> bash -c 'source <conda_sh> && <cmd>'`` invocation.
97
98         Lives beside (not on) DetectorRunner: a future Linux-native runner must not
99         inherit WSL plumbing. Requires ``self.cfg`` with ``distro``, ``conda_sh``
100         and ``timeout_sec`` attributes (TrackNetConfig and WasbConfig both qualify).
101         ``timeout_sec == 0`` maps to None = wait forever (configs default 1800s).
102         """
103
104         cfg: Any
105
106         def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
107             full = f"source {self.cfg.conda_sh} && {inner_cmd}"
108             argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
109             logger.debug("WSL exec: %s", full)
110             return subprocess.run(argv, capture_output=True, text=True,
111                                   timeout=(self.cfg.timeout_sec or None))
112
113         # ---- cache hygiene helpers (shared by both WSL runners) -----
114         def _wsl_free_gb(self, wsl_path: str) -> Optional[float]:
115             """Best-effort free space (GB) on the WSL filesystem holding ``wsl_path``.
116
117             Returns None if it can't be determined (never raises). Used as a pre-run
118             disk guard so a long extraction doesn't fill the disk mid-flight.
119             """
120             try:
121                 res = self._wsl_bash(f"df -P -B1 {wsl_tilde_quote(wsl_path)} | awk
122 'NR==2{{print $4}}'")
123             except (subprocess.TimeoutExpired, OSError):
124                 return None
125             try:
126                 return int((res.stdout or "").strip()) / (1024 ** 3)
127             except (ValueError, AttributeError):
128                 return None
129
130         def _wsl_rm_rf(self, wsl_paths: List[str]) -> None:
131             """Best-effort recursive delete of WSL paths (post-success cleanup; never
132             raises)."""
133             paths = [p for p in wsl_paths if p]
134             if not paths:
135                 return
136             quoted = " ".join(wsl_tilde_quote(p) for p in paths)
137             try:
138                 self._wsl_bash(f"rm -rf {quoted}")
139             except (subprocess.TimeoutExpired, OSError) as e:
140                 logger.warning("WSL cache cleanup failed (non-fatal): %s", e)
141
142         def wsl_source_unreadable_reason(self, src_mnt: str) -> Optional[str]:
143             """Actionable message if ``src_mnt`` is NOT readable from inside WSL, else None.
144
145             Google Drive (``G:``/Drive File Stream), UNC and many network/virtual
146             filesystems are invisible to WSL's ``/mnt`` ? staging a video from such a path
147             fails with a cryptic ``cp: cannot stat`` and the whole run silently produces no
148             trajectory (i.e. "0 rallies", a wasted run that looks like a quality result).

```

```

147         A cheap ``test -r`` preflight turns that into one clear, fixable error BEFORE
any
148         frame extraction. Best-effort: a flaky/timed-out probe returns None so the real
149         ``cp`` still gets its chance (we never block a genuinely-readable source).
150         """
151         try:
152             res = self._wsl_bash(f"test -r {shlex.quote(src_mnt)} && echo READABLE")
153         except (subprocess.TimeoutExpired, OSError):
154             return None
155         if "READABLE" in (res.stdout or ""):
156             return None
157         return (
158             f"video source is not readable from WSL at '{src_mnt}'. Google Drive, UNC, "
159             f"and other network/virtual filesystems are invisible to WSL's /mnt ? copy "
160             f"the video to a local disk (e.g. C:\\ or the WSL filesystem) and retry."
161         )
162
163
164     class DetectorRunner(ABC):
165         """Common interface for shuttle/ball trajectory detectors (WSL or otherwise)."""
166
167         @abstractmethod
168         def run_predict(self, video_win_path: str, output_win_dir: str,
169                        video_id: Optional[str] = None) -> Optional[str]:
170             """Run inference and return path to trajectory CSV (or None on failure).
171
172             ``video_id`` (the file MD5) namespaces all staged/cached/output artifacts so two
173             videos that happen to share a filename cannot reuse each other's frames or CSV.
174             It is optional for back-compat; callers in the live pipeline always pass it.
175             """
176             pass
177
178         @abstractmethod
179         def healthcheck(self) -> Tuple[bool, str]:
180             """Quick check if the runner (env, weights, etc.) is usable.
181
182             Returns (ok, message). Never raises. Cheap to call before a long run.
183             """
184             pass
185
186         # De facto trajectory contract: the trajectory hybrids call these
187         # unconditionally after run_predict. Concrete runners reuse/re-export the
188         # model-agnostic TrackNetRunner staticmethods (same CSV schema + windowing);
189         # a runner with custom windowing supplies its own with the same shapes.
190         parse_trajectory_csv: Callable[..., Any]
191         trajectory_to_action_windows: Callable[..., Any]
192

```

17. user (2026-06-18T07:44:02.898Z)

```

1     """WASB shuttle-trajectory inference on an arbitrary frame sequence (runs in WSL).
2
3     WASB-SBDT only ships *dataset* evaluation; this is a thin wrapper that reuses its
4     own building blocks (ImageDataset transform + detector + online tracker) to run on
5     any folder of extracted frames and emit a trajectory CSV in the exact shape the
6     indexer's `tracknet_runner.parse_trajectory_csv` already consumes:
7
8         Frame, Visibility, X, Y
9
10    It must run INSIDE the WSL `wasb` conda env, from the WASB-SBDT `src/` dir (it
11    imports WASB's `detectors`/`trackers`/`dataloaders`). The Windows-side adapter
12    (`wasb_runner.py`) stages this file + the frames into WSL and invokes it.
13
14    Resumable across reboots
15    -----

```

```

16 The GPU detector pass (one forward per sliding window ? ~21k for a 6-min clip) is
17 the slow, expensive part; the CPU tracker pass that turns detections into a
18 trajectory is cheap and *stateful* (it must see every frame in order, so it can't
19 resume mid-stream). We exploit that split:
20
21 * The detector pass writes its per-window raw outputs incrementally to
22 ``<cache>/det_raw.jsonl`` (flushed+fsync'd every ``--flush-every`` windows) and
23 records progress in ``<cache>/manifest.json`` (atomic write). A reboot loses at
24 most ``--flush-every`` windows of GPU work instead of the whole pass.
25 * On restart we replay the cached windows, skip the DataLoader ahead to the first
26 un-cached window, and continue. Once every window is cached, the detector pass
27 is skipped entirely and we just re-run the (cheap, deterministic) tracker over
28 the full ordered cache -> trajectory CSV. So a lost trajectory CSV costs seconds,
29 not the GPU hour.
30
31 The cache is keyed by the output path (``<out>_wasbcache/`` by default), lives next
32 to the output (NOT in %TEMP%), and is invalidated automatically if the frames dir,
33 weights, sport, window size, or frame count change.
34
35 Usage (in WSL, from ~/models/WASB-SBDT/src):
36 python wasb_infer.py --frames_dir /path/frames --weights
../pretrained_weights/wasb_badminton_best.pth.tar \
37 --out /path/traj.csv [--sport badminton] [--limit N] [--cache-dir DIR] [--flush-
every 1000] [--fresh]
38 """
39 import os
40 import os.path as osp
41 import csv
42 import glob
43 import json
44 import argparse
45 import logging
46 from collections import defaultdict
47 from contextlib import contextmanager
48
49 logger = logging.getLogger(__name__)
50
51 CACHE_VERSION = 1
52 _DET_FILE = "det_raw.jsonl"
53 _MANIFEST_FILE = "manifest.json"
54
55
56 def _build_cfg(weights: str, sport: str):
57     """Compose the WASB eval config via Hydra, overriding model/weights/device."""
58     from hydra import compose, initialize
59     with initialize(config_path="configs", version_base=None):
60         cfg = compose(config_name="eval", overrides=[
61             f"dataset={sport}",
62             "model=wasb",
63             f"detector.model_path={weights}",
64             "runner.device=cuda",
65             "runner.gpus=[0]", # this box has a single GPU; default config assumes
66             ])
67     return cfg
68
69
70 def _frame_id(path: str) -> int:

```

18. user (2026-06-18T07:44:03.336Z)

```

1 """Batch highlight-reel runner ? orchestrates "generate highlights for these videos
2 into this folder" requests with CLEAN, NON-DESTRUCTIVE, resumable execution.
3
4 why this exists (lessons paid for the hard way):

```

```

5 - **Never delete other work.** Reels + per-video run.logs are written to a LOCAL work
dir
6 first, then COPIED into the destination (atomically: temp name -> os.replace). The
7 destination folder is only ever ADDED to ? we never bulk-delete it. (A bulk `rm` of a
8 Google-Drive-synced folder once lost finished reels + logs.)
9 - **Drive-sync safe.** Generating directly inside a Google Drive folder races the sync
10 client (deleted files resurrect, partial writes sync). Generate locally; publish once.
11 - **Idempotent re-runs.** Each video's prior DB segments are cleared before processing
12 (add_video is an UPSERT and does NOT clear children), so a re-run never accumulates
13 duplicate rallies. Belt-and-suspenders with VideoCompiler's overlap-merge (PR #180).
14 - **Resumable & concurrency-friendly.** A video whose reel already exists in the
15 destination is skipped (unless --force); launching a second request never undoes the
16 first's published reels. The Gemini path also isolates its chunk workdir per video
17 (PR for output/chunks/<video_id>) so concurrent videos don't collide.
18
19 Usage:
20 python -m tools.generate_highlights --videos-dir "G:\\...\\June 12 2026" \
21 --out "G:\\...\\June 12 2026\\Highlights" --segmenter gemini [--force]
22 python -m tools.generate_highlights --video A.MP4 --video B.MP4 --out <dir> ...
23 """
24 from __future__ import annotations
25
26 import argparse
27 import logging
28 import os
29 import shutil
30 import sys
31 import traceback
32 from typing import Any, Dict, List, Optional, cast
33
34 from dotenv import load_dotenv
35
36 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
37
38 from backend.config import load_config
39 from backend.infrastructure.database import Database
40 from backend.pipeline.segmenters.base import segmenter_registry
41 from backend.pipeline.stitcher.compiler import VideoCompiler
42 from backend.results import Failure
43 from backend.utils.hashing import compute_video_id
44
45 logger = logging.getLogger("highlights_runner")
46
47 # Detectors that run in WSL can't read Google Drive / UNC via /mnt -> localize first.
48 _WSL_DETECTORS = {"wasb_hybrid", "tracknet_hybrid", "yolo_hybrid"}
49 _VIDEO_EXTS = (".mp4", ".MP4", ".mov", ".MOV", ".mkv", ".avi")
50
51
52 def _atomic_publish(local_path: str, dest_path: str) -> None:
53     """Copy local_path -> dest_path so the destination never sees a partial file.
54
55     Copy to a temp sibling, then os.replace (atomic on the same volume). Never deletes
56     anything else in the destination folder.
57     """
58     os.makedirs(os.path.dirname(dest_path), exist_ok=True)
59     tmp = dest_path + ".publishing.tmp"
60     shutil.copy2(local_path, tmp)
61     os.replace(tmp, dest_path)
62
63
64 def generate_highlights_batch(video_paths: List[str], out_dir: str, *, segmenter: str,
65                               local_work_dir: str, db_path: str, force: bool = False,
66                               skip_ai_handoff: bool = False, wasb_stream: bool = False,
67                               detect_from_paths: Optional[List[str]] = None) -> dict:
68     """Generate one highlight reel per input video into out_dir. Returns a summary dict.

```



```

69
70         Non-destructive: only ADDS reels/logs to out_dir (atomic publish); never deletes.
71
72         skip_ai_handoff: run the trajectory-hybrid segmenters fully-local (no Gemini
handoff);
73             candidate windows become the rallies. wasb_stream: have the WASB detector decode
the
74             video on the fly instead of dumping ~46k PNGs (output-preserving fast path,
#178).
75         detect_from_paths: optional list parallel to video_paths. When given, the detector
runs on
76             detect_from_paths[i] (a cheaper proxy ? e.g. a 1080p downscale of a 4K source)
but the
77             reel is stitched from video_paths[i] at full resolution. Rally timestamps
transfer 1:1
78             (a pure temporal-preserving downscale shares the timeline). None = detect+stitch
the
79             same source (legacy behaviour).
80         """

```

19. assistant (2026-06-18T07:44:13.068Z)

Let me check for any prior desktop/packaging mentions and the owned-model size tie-in, so the doc cross-links accurately and avoids duplication.

20. user (2026-06-18T07:44:15.314Z)

```

Found 13 files
docs\NEXT_STEPS.md
docs\CODE_MAP.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md
docs\PLATFORM_ARCHITECTURE.md
docs\BACKLOG.md
docs\COMMERCIALIZATION.md
docs\DEFERRED_HARDENING_PLAN.md
docs\VIDEO_LOCALITY_MODEL.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md

```

21. user (2026-06-18T07:44:16.525Z)

```

1:# Deferred Hardening Plan ? Train/Eval Guardrails, Async Jobs, Packaging (COMPLETED)
2-
3-**STATUS: COMPLETED** (2026-06-14). This planning doc's three items are all dispatched:
4-
5-- **#16 Async job model ? DONE** (PR #87; hardened by the self-scheduling drain loop in #149).
6:- **#7 Packaging modernization ? DONE** (PR #167: PEP 621 `pyproject.toml`, `setup.py` ?
`scripts/doctor.py`, `indexer` console entry point, CI switched to `pip install -e .[dev]`).
7-- **#13 Train/eval separation guardrails ? moved to [BACKLOG.md](BACKLOG.md)** (?; the
concrete `splits.py` / `assert_disjoint` match-level-split design is preserved there).
8-

```

22. user (2026-06-18T07:44:18.395Z)

```

36:weights); a *base* is the metal you forge. You pick one framework and feed it any clean base.
91: Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
shell (Swift/Kotlin/C++/Rust)
94:- Escalation order if a real Python CPU hot path ever appears: free-threading ?
numpy/vectorize ? Cython/PyO3 ? only then a

```

23. assistant (2026-06-18T07:44:26.646Z)

`docs/VIDEO\_LOCALITY\_MODEL.md` sounds directly on-thesis. Let me read it and the owned-model export snippet.

24. user (2026-06-18T07:44:27.711Z)

```
1      # Video locality ? where the bytes live (mental model + open questions)
2
3      **Status:** THINKING DOC (owner-requested, 2026-06-12) ? a shared mental model for
4      the upload/storage problem, with the open questions named. **No implementation is
5      greenlit by this doc** (per CLAUDE.md, both tracks stay owner-gated).
6      **Companions:** `UI_PROPOSAL.md` (B2/B3 upload/download), `HOSTING_PLAN.md`,
7      `PLATFORM_ARCHITECTURE.md` §3 (future StorageBackend/ComputeBackend registries),
8      collector ADR-002 (the frame-identical proxy contract this doc leans on).
9
10     ## 0. The questions that triggered this (and the short answers)
11
12     ***"Do we store the video they upload?"** Yes ? today the chunked-upload flow
13     assembles the full file under `security.upload_dir` (default `output/uploads`) on
14     our box, registers it by MD5, and nothing ever deletes it (retention policy is an
15     acknowledged gap ? BACKLOG). Processing also strews derived artifacts (chunks,
16     frames, trajectories, WSL temp) and, on the Gemini path, uploads ?1280px
17     re-encoded chunks to Google's Files API.
18
19     ***"How would they upload a 10 GB file?"** Today they can't: `/api/upload/init`
20     rejects anything over `security.max_upload_mb` (default **4096 MB**), and even
21     under the cap the file moves as sequential ?90 MB parts through a free-tier edge
22     whose body limit forced that part size. For scale: 10 GB ? 30?45 min of 4K phone
23     video (or ~2?3 h at 1080p); on a typical Indian uplink that's **~1 h at 20 Mbps,
24     ~4.5 h at 5 Mbps** ? before retries. Browser-tab uploads of that size are
25     miserable everywhere; expecting academies to manage S3/GCS buckets is a
26     non-starter. **The 10 GB problem is real and the current design doesn't solve it.**
27
28     ***"Can we make it so there is NO upload?"** Mostly ? because of $1.
29
30     ## 1. The central insight: timestamps are the product; the video is dead weight
31
32     Walk the pipeline and ask what each stage actually consumes:
33
34     - The **Gemini segmenter** never sees the original ? it re-encodes chunks down to
35       **1280 px** (`indexing.ai_chunk_scale_width`) before uploading them to Google.
36     - The **detectors/featurizer** consume downscaled frames and trajectories.
37     - The only consumer of the original full-res bytes is the **highlight compiler**,
38       and only to cut clips ? a pure-ffmpeg operation that can run **wherever the
39       original lives**.
40     - What the customer receives is `{rally intervals, report, reel}`. Intervals +
41       report are **kilobytes**. Given the intervals, the reel can be cut on the
42       machine that already has the original.
43
44     So: **any design that moves the 10 GB is moving the wrong thing.** The question
45     is never "how do we upload the video" but "what is the smallest artifact that
46     must cross the network for this tier of service."
47
48     ## 2. The locality ladder
49
50     From most to least bytes leaving the academy's Windows desktop. Sizes are rough
51     ratios for a 4K source ? measure on a real file before committing to any tier.
52
53     | | What crosses the network | ~Bytes (10 GB source) | What it needs | Status |
54     |---|---|---|---|---|
55     | **L3 ? full upload** (today's flow) | everything | 10 GB (blocked: 4 GB cap) | paid
56     edge tier + resumable uploads + server disk + retention policy | built, capped, wrong for big
57     files |
58
59     | **L2 ? proxy upload** | a timeline-identical ?1080p re-encode | ~1?2.5 GB | a tiny
60     client prep step (ffmpeg) ? **the exact frame-identical contract we already built and verified
61     as collector `prep-annotation-proxy`** (same fps, same frame count, parity-checked, MD5?MD5
```

manifest) | contract proven; client wrapper + server acceptance = new work |

57 | ****L1 ? chunks/candidates upload**** | only the ?1280px AI chunks (or just candidate windows) the Gemini path needs anyway | ~0.3?1 GB | client-side chunk+downscale (ffmpeg, CPU); server becomes a ****GPU-less orchestrator**** | the server half exists (it's the gemini segmenter's own flow, relocated) |

58 | ****L1-direct**** | chunks go ****straight from the client to Gemini**** ? zero video bytes through our server | ~0 through us | per-user key custody / metering / cost ceilings (OQ4) | idea only |

59 | ****L0 ? fully local**** | nothing (calls to Google only if the Gemini path is used) | 0 | the owned on-device model (roadmap north star) or a local GPU + our stack | not alpha-feasible; the long-term answer |

60 | ****L4 ? sneakernet**** | a hard drive | 0 over network | a person (the field associate / owner) | how the pilot actually works today |

61

62 ****The honest "no upload" answer:**** ***strictly*** no bytes leave only at L0 with the

63 owned model (and even L0 leaks ?1280px chunks to Google while the Gemini path is

64 the brain ? say that plainly in consent language, OQ3). The pragmatic near-term

65 answer is ****L2 or L1: don't eliminate the upload, shrink it 5?30x**** ? at which

66 point the existing 90 MB-part flow, the 4 GB cap, and a coffee-break wait are all

67 suddenly fine.

68

69 **## 3. Second-order effects worth noticing**

70

71 - ****L1/L2 make the serving box GPU-less for the Gemini tier**** ? ffmpeg + API

72 orchestration only. Hosting cost collapses; the GPU box stays for WASB/owned-

73 model work. (PLATFORM S3's ComputeBackend split falls out naturally.)

74 - ****Storing the proxy instead of the original**** is ~10x less disk AND a smaller

75 privacy surface; the review UI (alpha screen A4) actually ***prefers*** streaming a

76 1080p proxy to seeking inside a 10 GB original.

77 - ****The correction loop survives**** at every tier ?L1: labels are intervals on a

78 timeline, and the proxy timeline IS the original timeline by construction

79 (the collector ADR-002 invariants: same fps, same frame count, parity-verified).

80 - ****video_id keying:**** the indexer keys by MD5 of the file it processed. At L2

81 that's the proxy's MD5; the proxy-manifest pattern (original_md5 ? proxy_md5)

82 already exists ? but the product needs ONE registered identity rule (OQ2).

83

84 **## 4. The frugality frame ? afford the bills by not burning GPU**

85

86 The runway constraint (owner, 2026-06-12): ****the bills stay personally affordable**

87 for a long time as long as we don't burn too much GPU.****** The locality ladder maps

88 directly onto that:

89

90 - ****The Gemini serving path costs ~\$0.30?0.60 per hour of video**** in API calls

91 (2026-06-10 quality table) and at L1/L2 needs ****no GPU at all**** ? a CPU box,

92 ffmpeg, and an API key. At pilot scale (a handful of academies, a few sessions

93 a week) that is single-digit dollars a month: affordable indefinitely.

94 - ****GPU spend is the thing to guard.**** WASB / owned-model inference and training

95 stay on the owner's already-paid-for local 2070S; rent nothing until the owned

96 model has a ship-gate-passing reason to be served. L2/L1 turn this from a

97 discipline into a default: ***the serving box has no GPU to burn.***

98 - ****Storage follows the same logic:**** keep proxies, not originals (~10x less

99 disk); settle retention (OQ1) before volume arrives; the free edge tier stays

100 adequate precisely because L2 keeps payloads under the existing caps.

101 - ****Put an explicit cost ceiling per video**** (ties into OQ4): a 7-hour tape or a

102 runaway retry must not surprise-bill. The existing ops learnings (serial

103 uploads, `ai\_max\_workers=1`, bounded exponential retry) already point this way.

104

105 **## 5. The academy-visit dividend ? field work builds the product, not just the verdict**

106

107 The C13 visits aren't only a market read; run deliberately, each visit deposits

108 product assets:

109

110 - ****Phase-2 recordings (consented, adults-only) ARE the target corpus.**** The

111 owned model's ship target (0.66-to-beat) lives on exactly the messy multi-court

112 footage class these academies produce ? every show-back visit grows the

```

113     training set the model is starved for, at zero marginal engineering cost.
114 - **The show-back is a live dress rehearsal of the product loop:** footage comes
115     back on a drive (L4), we proxy + process + return a reel ? the same motion
116     L2 will automate. Every friction the associate hits there (file handoff,
117     proxy time, reel delivery) is a pre-paid product-backlog item; write it down.
118 - **Demo reactions and improvement asks** (captured verbatim on the answer
119     sheet) are the build backlog, collected by someone already standing in front
120     of the user.
121 - **Connectivity notes from visits** answer Q08 with data instead of guesses.
122
123     ## 6. Open questions (the decision backlog this doc exists to hold)
124
125     | # | Question | Why it matters / current lean |
126     |---|---|---|
127     | Q01 | **Retention policy per artifact class** ? original (only L3 has it), proxy,
chunks, frames/trajectories, Gemini remote files, intervals/labels | intervals/labels are the
asset (keep forever, tiny); originals/proxies need an explicit keep-while-active + delete-on-
request rule; Gemini remote-file GC already in BACKLOG |
128     | Q02 | **Which file's MD5 is the video's identity** at L2/L1 | lean: the processed
file's MD5 (proxy), with original_md5 carried in metadata ? the collector manifest pattern,
promoted to a product rule |
129     | Q03 | **Consent language for the Gemini path** ? even "local" modes ship 1280px chunks
to Google | must be plain in the academy-facing terms; paid-tier no-training posture to be cited
|
130     | Q04 | **Key custody for L1-direct** ? client calls Gemini with whose key? per-user
metering, abuse, cost ceiling per video | unsolved; this is what separates L1 from L1-direct |
131     | Q05 | **Windows client packaging** ? portable ffmpeg + a signed tiny app vs a
PowerShell script for the pilot | pilot can be a script the associate/owner runs; product needs
an installer story (ffmpeg LGPL build = licensing-clean) |
132     | Q06 | **What does the review UI play** at each tier | L2 streams the proxy (good); L1
has only chunks ? review degrades ? this quietly argues L2 over L1 as the default |
133     | Q07 | **Reel delivery** ? server-compiled from proxy (shareable 1080p) vs intervals-
back + client cuts full-res | could be both: proxy reel by default, "premium full-res" cut
locally |
134     | Q08 | **Real academy uplink numbers** ? the L2-vs-L3 case rests on bandwidth
assumptions | C13 visits can note connectivity informally; don't redesign around guesses |
135     | Q09 | **Recurring volume** ? an academy recording daily generates 10 GB/day; even L2
accumulates | retention (Q01) + maybe future delta thinking; not an alpha problem |
136     | Q010 | **If L3 must survive** (users who refuse any client-side step) | needs
resumable uploads + a paid edge tier + a raised cap; treat as the exception path, not the design
center |
137
138     ## 7. Where this lands (recommendation, not a greenlight)
139
140     **Default the alpha to L2 (proxy upload):** a one-command client prep step built
141     on the *already-verified* proxy contract, uploads that fit comfortably inside
142     today's chunked endpoint and 4 GB cap, a GPU-less serving path (the affordability
143     requirement, $4, satisfied by construction), a review UI that streams the proxy,
144     and intervals that map 1:1 onto the customer's original.
145     **L1/L1-direct is the fast-follow** once Q04 has an answer; **L0 is the owned-
146     model milestone**, not an alpha decision. The field pilot keeps running on L4 ?
147     a person with a hard drive is a fine storage backend for n=3 academies.
148

```

25. user (2026-06-18T07:44:28.376Z)

```

78     Apache/BSD shuttle/ball detectors for tennis/TT/pickleball/padel are not yet
identified.** Sequence by public-data
79     richness: **badminton ? tennis/TT ? pickleball/padel.**
80 - ? **Single-model-multisport-temporal-generalization is an OPEN HYPOTHESIS**, not
proven ? our corpus is badminton-only,
81     and the evidence once cited for it (RacketVision) was *refuted on verification* (it's
ball-tracking, not temporal). The
82     concept is sound on first principles (rally signal = a racquet-sport invariant) but
must be validated once a 2nd-sport

```

```

83     corpus exists. Don't cite RacketVision as evidence.
84
85     ## Language ? Python everywhere; on-device runtime is a deferred, layer-local export
    shell
86     - **Training/eval = Python-locked, and that's fine** (PyTorch/ms-swift/ActionFormer);
    offline/batch, no latency hot path.
87     - **Platform = no Python hot path to escape** ? the heavy bytes/GPU already run out-of-
    process (ffmpeg, WASB-in-WSL, cloud
88         GPU per-job); the control plane is I/O-bound. **Reject a Go/Rust hot-path split** ?
    single-digit-ms wins dwarfed by
89         ~40-min GPU jobs, and it would fracture the one codebase holding the QA-state + eval
    harness + provider registry.
90     - **On-device (deferred) = the only place a 2nd language appears, and it doesn't dictate
    our dev language:** train in
91         Python ? **export across the boundary** (ONNX/Core ML/ExecuTorch/GGUF) ? a thin device
    shell (Swift/Kotlin/C++/Rust)
92         runs the artifact with zero Python. **The one concrete constraint now: train with
    export-clean ops** so the compression
93         target stays reachable without re-architecting. (On-device base stays on the
    Qwen3-VL-4B Apache lineage ? *not* Gemma-3-270M.)
94     - Escalation order if a real Python CPU hot path ever appears: free-threading ?
    numpy/vectorize ? Cython/PyO3 ? only then a
95         separate service. **None exists today.**
96
97     ---
98
99     ## Phase 0 ? Foundation (start now, per-milestone greenlight; $0, local, no cloud/DPA)
100
101     ### M0.1 ? The labeled-corpus spine **+ the event-index contract** (the moat) .
    *greenlight item*
102     Canonical JSONL row per rally (corpus) **and** the queryable per-video event-index
    schema (the conversational seam)
103     ? same row, wired to a store. Fields: `{video_id(MD5), rally_ordinal, sport, start, end,
    ending_reason/fault_tags,
104     shot_count, label_type, source(human|pseudo|vendor), consent/training_opt_in, split(BY
    MATCH), trajectory_ref,
105     labeled_window, derived_clip_ref, confidence}`. 3 transcoders (heuristic/LogReg · TAL
    ActivityNet-JSON · VLM clip?QA),
106     one scorer spine (`metrics.match_intervals`). Authored in `sports-data-collector`
    (system-of-record; extends PR #13).
107     **Reserve fault/event taxonomy now.** Acceptance: round-trip test; split-by-match
    enforced; **no Gemini-sourced rows in any training view**.

```

26. assistant (2026-06-18T07:45:22.695Z)

I have everything I need. `VIDEO\_LOCALITY\_MODEL.md` (its L0 "fully local" tier + OQ5 packaging) and the owned-model plan's "export across the boundary" section are the perfect anchors, and `backend/pipeline/detectors/base.py` literally pre-specifies the de-WSL native runner. Let me set up the branch off `origin/master`.

27. user (2026-06-18T07:45:23.758Z)

```

branch 'docs/desktop-app-plan' set up to track 'origin/master'.
Switched to a new branch 'docs/desktop-app-plan'
---VERIFY---
docs/desktop-app-plan
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)

```

28. user (2026-06-18T07:45:30.106Z)

Chapter marked: "Desktop app plan doc". Continue your current work.

29. assistant (2026-06-18T07:45:29.419Z)

Now I'll write the tracked project doc.

30. assistant (2026-06-18T07:47:44.579Z)

Cross-Platform Desktop App ? "Plug in your card, get your highlights"

```
> **Status:** `DRAFT` ? `owner-gated` (scoping doc only; **nothing is built**) . **Owner:**
avidullu . **Created:** 2026-06-18 . **Last updated:** 2026-06-18
> **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** $7 progress tracker is the source of truth; indexed in `docs/README.md`;
pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once the owner greenlights P0).
> **Relation to existing docs:** **peer-of**
[ `PLATFORM_ARCHITECTURE.md` ] (PLATFORM_ARCHITECTURE.md) (this is the **`LocalDesktop`
ComputeBackend** made into a product) . **realizes the L0 tier** of
[ `VIDEO_LOCALITY_MODEL.md` ] (VIDEO_LOCALITY_MODEL.md) . **consumes**
[ `OWNED_MODEL_IMPLEMENTATION_PLAN.md` ] (OWNED_MODEL_IMPLEMENTATION_PLAN.md) (the smaller/export-
clean model is one answer to the compute crux) . bound by
[ `COMMERCIALIZATION.md` ] (COMMERCIALIZATION.md) licensing guardrails.
> **Honesty note:** every claim is tagged `[verified]` (checked against code/measured),
`[researched]` (needs sign-off), or `[design]` (proposed). **The entire build is `[design]`** ?
this doc scopes; it does not ship. Not legal advice.
```

0. TL;DR

Package the mature local pipeline (WASB shuttle detector + improved windowing + ffmpeg stitch, already runnable fully-offline via [`tools/generate\_highlights.py --skip-ai-handoff`] (./tools/generate_highlights.py)) as a ****cross-platform desktop app**** for ****non-technical recreational players****: plug in the camera's USB/SD card, click once, get back ****only the highlight reels**** ? fully on-device, ****no upload****, ****no copy of the GBs of 4K originals****. This is the product surface that fits the "local, cheap, efficient" thesis and the L0 ("nothing leaves the machine") tier of `VIDEO\_LOCALITY\_MODEL.md`.

****The make-or-break unknown is compute on a typical enthusiast laptop**** ? which often has ****no discrete GPU****. WASB runs at ****~3.4x real-time on a laptop RTX 2070 Super**** `[verified, measured]`; on CPU it is far slower, and Apple Silicon has ****no CUDA**** at all. ****P0 is a feasibility spike that measures CPU and Apple-MPS throughput for a ~10-min clip before any app is built.**** If CPU-only is unacceptable on commodity hardware, the path needs a smaller/faster owned model (ties `OWNED\_MODEL\_IMPLEMENTATION\_PLAN`) or a tiered "fast-if-GPU, slow-but-works-if-CPU" design. ****Everything past P0 is gated on that result.****

1. Problem & goal

****Why now / the user.**** Target users are ****recreational sports enthusiasts, not tech-savvy****. Their match videos are ****huge**** (GBs of 4K). The cloud-upload product surface fails them on every axis: slow on a typical Indian uplink (`VIDEO\_LOCALITY\_MODEL.md` \$0: 10 GB ? ~1 h at 20 Mbps, ~4.5 h at 5 Mbps), bandwidth-costly, privacy-fraught, and it bloats both our storage and their drives. The thing the customer actually wants back is ****kilobytes of timestamps ? a highlight reel**** (`VIDEO\_LOCALITY\_MODEL.md` \$1: ****"timestamps are the product; the video is dead weight"**) .**

****The concrete outcome ("good" looks like this):****

1. User plugs the GoPro/phone SD card or camera USB into their laptop.
2. The app detects the card, lists the match videos on it.
3. User clicks ****Generate Highlights****.
4. The app makes an on-device proxy, runs WASB detection + windowing + stitches the reel ? ****100% locally****.
5. The reel(s) land in a user-chosen folder. ****The originals are never copied off the card and never bloat any drive.****

No account, no upload, no cloud dependency in the default path. Gemini stays an ****optional**** "better brain if you want it" toggle ? never required, never bundled as a hard dependency, never a training source (`COMMERCIALIZATION` C4).

****Read to get current:**** ``VIDEO_LOCALITY_MODEL.md`` (the L0/locality framing + OQ5 packaging seed), ``PLATFORM_ARCHITECTURE.md`` S3.2 (the ComputeBackend seam), `[`backend/pipeline/detectors/base.py`](../backend/pipeline/detectors/base.py)` (the detector ABC that already anticipates a Linux-native runner), ``tools/generate_highlights.py`` (the local entrypoint this app wraps).

2. Decisions locked

#	Decision	Source / date	Implication
D1	**Default path is 100% on-device, no upload, no Gemini.** The reel is produced from the local pipeline (<code>`--skip-ai-handoff`</code>). This doc · 2026-06-18; matches <code>`VIDEO_LOCALITY_MODEL`</code> L0 The app must never *require* a network call. Gemini is an opt-in toggle only.		
D2	**Never copy the originals.** The app reads source video in place (off the card / a chosen folder) and writes **only** reels + a tiny report to a user folder. This doc; mirrors <code>`generate_highlights.py`</code> non-destructive <code>`_atomic_publish`</code> discipline No "import library" that duplicates 10 GB files. Detection runs on a proxy; stitch reads the original in place.		
D3	**Drop the WSL2 dependency; run the detector natively per-OS.** This doc; enabled by the <code>`DetectorRunner`</code> ABC <code>`[verified]`</code> Windows-native (no WSL), Linux+NVIDIA (CUDA directly), macOS (MPS or CPU). The WSL adapter stays for the current dev box but is **not** the shipped path.		
D4	**Commercial-clean licensing for everything bundled: MIT / Apache-2.0 / BSD only** (code, weights). ffmpeg ships as an **LGPL build** (no GPL-only encoders like x264). <code>`COMMERCIALIZATION`</code> (ratified 2026-06-09); <code>`VIDEO_LOCALITY_MODEL`</code> OQ5 (**ffmpeg LGPL build = licensing-clean**) WASB-SBDT weights are MIT <code>`[verified]`</code> (see <code>`tracknet-wasb-wsl2-decision`</code>). The stitch must use openh264 (BSD) or OS hardware encoders, **not** statically-linked x264. See S3.4.		
D5	**Reuse the existing pipeline verbatim; add only the shell + a native runner + device-IO.** <code>`backend/pipeline/detectors/base.py`</code> reuse contract <code>`[verified]`</code> No re-architecture of windowing/stitch. New code = native runner, USB/SD detection, app shell, packaging.		
D6	**Feasibility is gated on P0.** No app shell, no packaging work starts until the compute spike returns a verdict. This doc; <code>`decision-rigor-norm`</code> (NUMBERS INFORM, FOUNDATIONS DECIDE) P0 is the only un-gated deliverable; P1?P3 are owner-greenlit *after* P0 numbers.		

3. Foundation ? research / prior art ``[design]`` / ``[researched]``

3.1 The compute reality (the crux) ? what we know vs must measure

- ****WASB on a discrete laptop GPU is acceptable**** ``[verified, measured]``: on the owner's ****RTX 2070 Super Max-Q (8 GB)****, WASB inference is the pipeline driver at ****~3.2?3.4× real-time**** (23.4 GPU-min per footage-min, ~0.055 s/frame), peak VRAM ~5.7?6.0 GB; proxy (NVENC) ~0.2× RT; 4K stitch (CPU libx264) ~16 s/rally; end-to-end 25× clip length. Source: ``current-state.md`` ``COST_SCALING.md`` checkpoint (6 Boxhill clips, 2026-06-18).
- ****The typical enthusiast laptop has NO discrete GPU.**** A 10-min 4K clip at 3.4× RT is ~34 GPU-min ***with*** a 2070S. On integrated graphics / CPU-only the wall-clock is unknown and could be 10×+ worse ? possibly ****hours per clip****, which would kill the one-click promise.
- ****Apple Silicon has no CUDA.**** WASB/TrackNet weights are PyTorch; PyTorch MPS (Metal) backend ***may*** run them, but op-coverage and speed on the WASB model are ****unverified**** ``[design]``. CPU fallback on Apple is also unverified.
- ****Owned-model tie-in.**** ``OWNED_MODEL_IMPLEMENTATION_PLAN.md`` already states the on-device answer: ***train in Python ? **export across the boundary**** (ONNX / Core ML / ExecuTorch / GGUF) ? a thin device shell runs the artifact with zero Python*, with the standing constraint to ****train with export-clean ops****. A smaller, exported, quantized shuttle detector is a credible answer if the CPU/MPS numbers on the current WASB weights are bad. ****This makes P0 partly a referendum on whether the desktop product needs the owned-model track.****

3.2 The de-WSL native runner is already designed-for ``[verified]``

The detector layer was deliberately built to make this port low-regret. `[`backend/pipeline/detectors/base.py`](../backend/pipeline/detectors/base.py)` documents, verbatim, an ****"Optional Linux-native path (the main de-risk goal)"****: a ``LinuxNativeRunner``/``ONNXRunner`` that ****"Skip[s] all wsl / _stage / conda activate, load[s] weights locally (torch or onnx)? write[s] the exact same CSV format so ``trajectory_to_action_windows`` works verbatim."**** The WSL plumbing lives in a separate ``WslCommandMixin``, explicitly so a native runner ****"must not inherit WSL plumbing."**** The actual inference math is already isolated in

```
[`wasb_infer.py`](../backend/pipeline/detectors/wasb_infer.py) (torch + the WASB-SBDT modules, emitting `Frame,Visibility,X,Y`). **So the port is: lift `wasb_infer.py` out of WSL, load weights with native torch/onnxruntime per-OS, keep the identical CSV contract.** The windowing brain (`trajectory_to_action_windows`) and stitch are untouched.
```

3.3 Compute backends to evaluate in P0

```
| Path | OS | Toolkit | Status |
|---|---|---|---|
| CUDA-native | Windows-native, Linux+NVIDIA | torch+CUDA (no WSL) | proven in WSL `[verified]`; native = port |
| Apple MPS | macOS (Apple Silicon) | torch MPS / Core ML export | **unverified ? P0 must measure** |
| CPU | all (the no-GPU laptop) | torch-CPU / **onnxruntime** (often fastest on CPU) | **unverified ? P0 must measure; the gating number** |
| Smaller/exported model | all | ONNX/Core ML/ExecuTorch, quantized | deferred to owned-model track; P0 informs whether it's *needed* |
```

3.4 App-shell prior art & licensing landscape `[researched]`

- **App shell options:** **Tauri** (Rust + system webview; ~3?10 MB binaries; MIT/Apache) vs **Electron** (bundled Chromium; ~100?150 MB; MIT) vs **CLI + installer** (wrap `generate\_highlights.py`; smallest effort). All shell licenses are clean.
- **Backend packaging:** the Python/FastAPI backend bundles via **PyInstaller** or a frozen venv as a local "sidecar." Packaging was already modernized ? **PEP 621 `pyproject.toml` + an `indexer` console entry point** shipped (`DEFERRED\_HARDENING\_PLAN` #7, DONE, PR #167) `[verified]` ? so a CLI MVP has a real head start.
- **ffmpeg licensing (a genuine landmine):** ffmpeg is LGPL/GPL. The current stitch path uses **libx264 (GPL)**. Redistributing that inside an installer would impose GPL on the bundle. **Mitigation (D4):** ship an **LGPL ffmpeg build** using **openh264 (BSD)** or the OS hardware encoder (NVENC / Apple **VideoToolbox** / Linux **VAAPI**) ? the desktop app should prefer hardware H.264 anyway (faster, no CPU stitch tax).
- **Weights:** WASB-SBDT = MIT `[verified]` (`tracknet-wasb-wsl2-decision`), redistributable.

4. Design / architecture `[design]`

4.1 The one-click flow (and what code each step reuses)

```
...
[card inserted]
  ? (NEW) per-OS device-insert watcher ? mount path
  ?
[list videos on the card]  ? glob video exts; reuse _VIDEO_EXTS from generate_highlights.py
  ?
[make on-device proxy]    ? ffmpeg downscale (hardware encoder); reuse the proxy + --detect-
from split (#205)
  ?
[detect: WASB native]     ? (NEW) NativeWasbRunner (DetectorRunner subclass) ? NO WSL; same
CSV contract
  ?
[windowing ? rallies]     ? REUSE trajectory_to_action_windows + improved milestone windowing
(R1: cap=6 + R4)
  ?
[stitch reel from ORIGINAL] ? REUSE VideoCompiler (backend/pipeline/stitcher/compiler.py); read
original in place
  ?
[export reel + tiny report to a user folder] ? reuse _atomic_publish non-destructive write;
NEVER copy originals
...
```

The detector runs on a **proxy**; the stitch reads the **original on the card** (the `--detect-from` split, already shipped `[verified]` in `generate\_highlights.py`). Rally timestamps transfer 1:1 because the proxy shares the timeline (the collector frame-identical proxy contract, `VIDEO\_LOCALITY\_MODEL` §1).

4.2 Reuse map ? what's NEW vs REUSED

```
| Concern | Status | Code |
|---|---|---|
```



```
| Trajectory?rally windowing | **REUSE** `[verified]` |
`tracknet_runner.trajectory_to_action_windows` (model-agnostic staticmethod) |
| Stitch / reel compile | **REUSE** `[verified]` |
`backend/pipeline/stitcher/compiler.py::VideoCompiler` (pure ffmpeg) |
| Local orchestration (proxy?detect?stitch, non-destructive, resumable, `--skip-ai-handoff`,
`--detect-from`) | **REUSE** `[verified]` | `tools/generate_highlights.py` |
| Detector runner contract | **REUSE** `[verified]` |
`backend/pipeline/detectors/base.py::DetectorRunner` ABC |
| WASB inference math | **REUSE** (lift out of WSL)** `[verified]` |
`backend/pipeline/detectors/wasb_infer.py` |
| **Native (de-WSL) WASB runner, per-OS** | **NEW** `[design]` | `NativeWasbRunner` ? sibling of
`wasb_runner.py`, no `WslCommandMixin` |
| **USB/SD insert detection + video listing** | **NEW** `[design]` | per-OS: Win
WMI/`WM_DEVICECHANGE`, macOS DiskArbitration, Linux udev/udisks2 |
| **App shell / one-click UI** | **NEW** `[design]` | Tauri vs Electron vs CLI (§3.4; P2
decides) |
| **Packaging + code-signing per OS** | **NEW** `[design]` | PyInstaller/MSIX (Win),
.dmg+notarize (mac), AppImage/Flatpak (Linux) ? P3 |
```

4.3 Where this sits in the platform architecture

This **is** the `local` / `LocalDesktop` **ComputeBackend** of `PLATFORM\_ARCHITECTURE.md` §3.2, realized as a shippable product instead of a server worker. The control-plane/data-plane split degenerates cleanly to "all on one machine." Nothing here forecloses the hosted SaaS path ? it's the privacy-max end of the same spectrum, and a genuine differentiator (no competitor offers true on-device compute except SwingVision, which proves it's commercially viable ? `PLATFORM\_ARCHITECTURE` §intro).

4.4 App-shell recommendation (to confirm at P2)

Lean: **start with a CLI + installer MVP** (wraps the existing `indexer`/`generate\_highlights.py` entry point ? fastest way to validate packaging, signing, USB-detect, and the native runner end-to-end on all three OSes), then **graduate to a Tauri GUI** for the non-technical user (tiny download fits the "huge-video laptops, slim app" frame; the Python backend runs as a sidecar the Tauri front-end calls over localhost). Electron is the fallback if webview inconsistencies bite. This mirrors the repo's "simplest thing first, graduate on evidence" bias.

5. Risk table `[design]`

Risk	Severity	Mitigation / where decided
---	---	---
CPU/MPS too slow on a no-GPU laptop (kills the one-click promise)	make-or-break	P0 spike measures it before anything is built (§6, §7-P0). Fallbacks: tiered fast/slow UX, or the owned smaller model.
ffmpeg GPL contamination via libx264 in the installer	high (legal)	D4: LGPL build +openh264/hardware encoders only.
Code-signing friction (mac notarization mandatory; Win SmartScreen without EV cert)	medium	P3 scopes certs/notarization per OS; budget the Apple Developer Program + a Windows cert.
Privacy regressions (the on-device promise is the selling point)	medium	D1/D2: no upload, no original-copy, no required network. Make the "nothing leaves your machine" guarantee testable (assert no outbound calls in default mode).
Per-OS device-insert flakiness / removable-media edge cases (card pulled mid-run)	medium	P1/P2 handle gracefully; never write to the card; fail safe.
Support burden for non-technical users (drivers, "it didn't detect my card")	medium	favors a polished GUI + clear errors (reuse the actionable-error discipline already in the WSL adapter).

6. Honest limits ? what this does NOT do

- **A DRAFT/DONE status here proves nothing about runtime feasibility.** The doc is a scope; only the P0 numbers decide whether the product is buildable on commodity hardware.
- **If CPU-only is too slow on a typical laptop, the simple "wrap the pipeline" path does not exist** ? it becomes "ship a smaller owned model" (a much larger effort, ties `OWNED\_MODEL\_IMPLEMENTATION\_PLAN`) or "GPU-tier only" (shrinks the addressable market). The doc names this honestly rather than assuming the wrap is free.
- **No accuracy claim changes.** The desktop app inherits exactly the current rally-detection quality (R1 milestone: precision is still the open gap per `RALLY\_DETECTION\_GAP\_CLOSURE.md`);

packaging does not improve detection.

- **Gemini-quality reels need the optional online toggle.** The fully-local default uses `--skip-ai-handoff`; users wanting the Gemini brain accept that 1280px chunks leave the machine (consent language, `VIDEO_LOCALITY_MODEL` OQ3`).
- **macOS support is contingent on P0's MPS/CPU verdict** ? it is not assumed.

7. Deliverables & progress tracker ? **source of truth**

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands. **Everything below is owner-gated; only this doc (the scoping PR) is in flight today.**

ID	Deliverable	Depends on	Gated?	Status	PR
D0	This scoping doc (<code>DESKTOP_APP_PLAN.md` + README index</code>)			No	?(this PR)
P0	Compute-feasibility spike (make-or-break). Measure WASB throughput for a ~10-min clip on: (a) CPU-only (torch-CPU & onnxruntime), (b) Apple MPS, (c) confirm CUDA-native (no WSL). Output: a wall-clock table + a go/no-go verdict + "do we need a smaller model?" recommendation. No app code.	D0	Owner greenlight	?	?
P1	De-WSL native detector port (<code>NativeWasbRunner`</code> , per-OS): CUDA-native (Win/Linux), MPS/CPU fallback, identical <code>Frame,Visibility,X,Y`</code> CSV; reuse windowing verbatim; healthcheck per backend.	P0 verdict	Yes	?	?
P2	Desktop app shell + one-click flow : USB/SD detect ? list ? proxy ? native detect ? stitch ? export; shell choice (CLI MVP ? Tauri) confirmed; "no upload / no original-copy" enforced & testable.	P1	Yes	?	?
P3	Packaging + code-signing installers per OS (Win MSIX/Authenticode, mac .dmg + Developer-ID notarization, Linux AppImage/Flatpak); LGPL ffmpeg bundling; MIT/Apache/BSD audit of the bundle.	P2	Yes	?	?

8. Open questions ? owner / external

Blocking gates (owner decides before the dependent phase):

- (gates P1?P3) Greenlight P0** ? the spike is cheap and decides everything. Recommend: yes, run P0 next.
- (gates P1) If CPU is too slow, which fork?** tiered GPU-fast/CPU-slow UX · invest in a smaller exported owned model (`OWNED_MODEL_IMPLEMENTATION_PLAN``) · GPU-tier-only product (narrower market).
- (gates P2) App shell:** confirm CLI-MVP-then-Tauri, or go straight to a GUI / Electron.
- (gates P3) Distribution & signing budget:** Apple Developer Program (mandatory for notarization), a Windows code-signing cert (EV for clean SmartScreen) ? owner cost decision.

Nice-to-knows:

- OS coverage priority for v1 (Windows-first? given the owner's box and the academy beachhead).
- Auto-update strategy (Tauri updater / Sparkle / none for the pilot).
- Does the desktop app reuse the optional Gemini toggle at all, or stay strictly local for v1?

9. Definition of done

This subproject is DONE (? flip Status, archive) when:

- [] **P0** produced a measured throughput table (CPU / MPS / CUDA-native) and an owner-accepted go/no-go verdict (incl. whether a smaller model is required). *(P0 alone is a valid stopping point if the verdict is "not feasible without the owned model" ? then this doc hands off to `OWNED_MODEL_IMPLEMENTATION_PLAN``.)
- [] **P1** the native (de-WSL) detector runs on ?1 GPU OS and ?1 CPU/MPS path, producing byte-compatible trajectories (regression vs the WSL runner on a shared clip).
- [] **P2** a non-technical user can insert a card and get a reel with no terminal, no upload, and no copied original ? verified end-to-end on ?1 OS.
- [] **P3** signed/notarized installers exist for the greenlit OSes; the bundle passes an MIT/Apache/BSD (+ LGPL-ffmpeg) license audit.
- [] \$7 rows all ? and the headline feasibility verdict is recorded for posterity.

10. References

Internal code anchors `[verified]`:

[`tools/generate_highlights.py``](`./tools/generate_highlights.py`) (local orchestration; `--skip-ai-handoff``, `--wasb-stream``, `--detect-from``, `_WSL_DETECTOR``, `_atomic_publish``) ·

```
[`backend/pipeline/detectors/base.py`](../backend/pipeline/detectors/base.py) (`DetectorRunner`
ABC + the documented Linux-native de-risk path + `WslCommandMixin`) .
[`backend/pipeline/detectors/wasb_runner.py`](../backend/pipeline/detectors/wasb_runner.py) (the
WSL adapter to replace) .
[`backend/pipeline/detectors/wasb_infer.py`](../backend/pipeline/detectors/wasb_infer.py) (the
torch inference core to lift native) . [backend/pipeline/detectors/tracknet_runner.py`](../back
end/pipeline/detectors/tracknet_runner.py) (shared `trajectory_to_action_windows`) .
[`backend/pipeline/segmenters/wasb_hybrid.py`](../backend/pipeline/segmenters/wasb_hybrid.py) .
`backend/pipeline/stitcher/compiler.py::VideoCompiler` . `pyproject.toml` + `indexer` console
entry (#167).
**Internal docs:** [VIDEO_LOCALITY_MODEL.md](VIDEO_LOCALITY_MODEL.md) (L0 fully-local tier;
OQ5 packaging) . [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) §3.2 (ComputeBackend
seam) . [OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md) (export-clean
on-device model) . [COMMERCIALIZATION.md](COMMERCIALIZATION.md) (C4 paid-LLM no-train;
MIT/Apache/BSD-only) . [DEFERRED_HARDENING_PLAN.md](DEFERRED_HARDENING_PLAN.md) (#7 packaging,
DONE) . [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md) (the WSL setup being superseded for
shipped builds).
**Measured source:** `current-state.md` 2026-06-18 `COST_SCALING.md` checkpoint (WASB ~3.4× RT
on RTX 2070 Super Max-Q, peak VRAM ~5.7?6.0 GB).
**External (to validate at P0/P2/P3):** PyTorch MPS backend . onnxruntime (CPU/CoreML/DirectML
execution providers) . Tauri vs Electron sizing . ffmpeg LGPL build /openh264 . Apple
Developer-ID notarization . Windows Authenticode/MSIX.
```

Changelog

- `2026-06-18` ? Created. Scoping doc for the cross-platform desktop app + the de-WSL native-compute port; all `[design]`. §7 P0 (compute-feasibility spike) named the make-or-break gate. Owner-gated; nothing built.

31. user (2026-06-18T07:47:44.609Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\docs\DESKTOP_APP_PLAN.md (file state is current in your context ? no need to Read it back)

32. assistant (2026-06-18T07:48:13.140Z)

The doc is written. Now let me index it in `docs/README.md`. The top block is the rally-quality track (no header); a desktop/product scoping doc fits best under a new section. Let me add it before "Practical / ongoing".

33. user (2026-06-18T07:48:19.106Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

34. assistant (2026-06-18T07:48:23.445Z)

Both files are ready. Let me re-verify the branch (per branching-safety) and commit with explicit paths only.

35. user (2026-06-18T07:48:25.554Z)

```
---BRANCH---
docs/desktop-app-plan
---HEAD (should be mine/origin master tip)---
a010667 feat(eval): nightly all-OFF floor ? pin a baseline_off reference config (#217)
---ONLY MY COMMITS vs origin/master (should be empty pre-commit)---
---STATUS---
M backend/eval/rally_seg_eval.py
M docs/README.md
?? artifacts/
?? docs/DESKTOP_APP_PLAN.md
```

?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md

36. assistant (2026-06-18T07:48:39.490Z)

I notice `backend/eval/rally\_seg\_eval.py` is modified ? I did **not** touch it; it's sibling/owner WIP that appeared in the tree (exactly the shared-checkout scenario the convention warns about). I'll stage **only my two explicit paths** and leave everything else untouched.

37. user (2026-06-18T07:48:41.121Z)

warning: in the working copy of 'docs/DESKTOP_APP_PLAN.md', LF will be replaced by CRLF the next time Git touches it

---STAGED (should be ONLY my 2 files)---

docs/DESKTOP_APP_PLAN.md

docs/README.md

---README staged diff (confirm only my added section)---

diff --git a/docs/README.md b/docs/README.md

index c2e7266..b9b9ca2 100644

--- a/docs/README.md

+++ b/docs/README.md

@@ -10,6 +10,9 @@

- GOLDEN_REGRESSION_FIXTURES.md ? **Design** (owner-gated, NOT started): **video-free**, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **\$7 progress tracker + definition-of-done**. Extends GOLDEN_DATA_SHARING.md.

+++ Product / platform plans

+ DESKTOP_APP_PLAN.md ? **Tracked** scoping doc (2026-06-18, owner-gated, `design` ? **NOTHING** built): **package** the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical **no-GPU** enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? \$7 tracker = **P0 compute-feasibility spike** ? P1 native detector ? P2 app shell ? P3 packaging/installers. Realizes the **fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

+

Practical / ongoing

- TRACKNET_WSL_SETUP.md

- Golden set docs (`GOLDEN\_SET\_\*`)

38. assistant (2026-06-18T07:48:44.904Z)

Staged content is exactly my two files. Committing and pushing.

39. user (2026-06-18T07:48:53.092Z)

Exit code 2

[docs/desktop-app-plan f721274] @ docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port

2 files changed, 167 insertions(+)

create mode 100644 docs/DESKTOP_APP_PLAN.md

/usr/bin/bash: eval: line 33: syntax error near unexpected token `('

/usr/bin/bash: eval: line 33: `USB/SD card, click once, get back ONLY the highlight reels, fully on-device (no'

40. assistant (2026-06-18T07:49:02.542Z)

The commit landed but with a mangled message ? I used PowerShell here-string syntax (`@'...'@`) inside the ****Bash**** tool (POSIX sh), so the `@` became literal and the parens broke parsing. Let me check and fix the commit message.

41. user (2026-06-18T07:49:03.536Z)

f721274 @ docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
---FULL MSG---

@
docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port

New tracked project doc DESKTOP_APP_PLAN.md (owner-gated, all [design] ? nothing built). Vision: package the mature local pipeline (WASB + improved windowing + ffmpeg stitch, already runnable via `generate\_highlights.py --skip-ai-handoff`) as a desktop app for non-technical recreational players ? plug in the cameras

42. assistant (2026-06-18T07:49:09.753Z)

I'll write the commit message to a temp file and amend with `-F` (clean, no shell-quoting pitfalls).

43. assistant (2026-06-18T07:49:15.519Z)

docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port

New tracked project doc DESKTOP_APP_PLAN.md (owner-gated, all [design] ? nothing built). Vision: package the mature local pipeline (WASB + improved windowing + ffmpeg stitch, already runnable via generate_highlights.py --skip-ai-handoff) as a desktop app for non-technical recreational players ? plug in the camera's USB/SD card, click once, get back ONLY the highlight reels, fully on-device: no upload, no original-video bloat.

Centerpiece = the de-WSL native-compute port (run WASB natively per-OS: Linux/Windows CUDA, macOS MPS/CPU, no WSL2), grounded in the DetectorRunner ABC that already anticipates a Linux-native runner. The make-or-break unknown is compute on a typical NO-GPU enthusiast laptop (WASB ~3.4x real-time on a laptop 2070S); the doc names a P0 feasibility spike to measure CPU/MPS before any app is built.

Follows PROJECT_DOC_TEMPLATE (status header, decisions-locked, section-7 tracker as source-of-truth: P0 spike -> P1 native detector -> P2 app shell -> P3 packaging, definition-of-done). Realizes the L0 tier of VIDEO_LOCALITY_MODEL and the LocalDesktop ComputeBackend of PLATFORM_ARCHITECTURE section 3.2; ties the smaller export-clean owned model as one answer to the compute crux. Indexed in docs/README.md. Owner-gated implementation ? only this scoping doc is in flight.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

44. user (2026-06-18T07:49:15.543Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch_desktop_commit_msg.txt (file state is current in your context ? no need to Read it back)

45. user (2026-06-18T07:49:22.487Z)

[docs/desktop-app-plan b9ff235] docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port

Date: Thu Jun 18 13:18:53 2026 +0530
2 files changed, 167 insertions(+)
create mode 100644 docs/DESKTOP_APP_PLAN.md
---AMENDED SUBJECT---

```
b9ff235 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
---VERIFY ONLY MY COMMIT vs origin/master---
b9ff235 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
---STILL ONLY 1 FILE-SET CHANGED---
b9ff235 docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
docs/DESKTOP_APP_PLAN.md | 164 ++++++
docs/README.md           | 3 +
2 files changed, 167 insertions(+)
```

46. assistant (2026-06-18T07:49:26.930Z)

Clean ? only my commit, only my two files. Now push and open the PR.

47. user (2026-06-18T07:49:29.691Z)

```
remote:
remote: Create a pull request for 'docs/desktop-app-plan' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/desktop-app-plan
remote:
branch 'docs/desktop-app-plan' set up to track 'origin/docs/desktop-app-plan'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/desktop-app-plan -> docs/desktop-app-plan
```

48. user (2026-06-18T07:49:44.702Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch_desktop_pr_body.md (file state is current in your context ? no need to Read it back)

49. assistant (2026-06-18T07:49:44.683Z)

What

Stands up a **new tracked subproject** as a scoping doc: `docs/DESKTOP\_APP\_PLAN.md` (+ one-line index in `docs/README.md`). **Owner-gated**, everything `[design]` ? nothing is built. **This is for future sessions to build on.**

Vision

Package the mature **local** pipeline (WASB shuttle detector + improved windowing + ffmpeg stitch ? already runnable fully-offline via `tools/generate\_highlights.py --skip-ai-handoff`) as a **cross-platform desktop app** for **non-technical recreational players**:

> plug in the camera's USB/SD card ? click once ? get back **only the highlight reels** ? fully on-device, **no upload**, **no copy of the GBs of 4K originals**.

This is the product surface that fits the "local, cheap, efficient" thesis and the **L0** (nothing leaves the machine) tier of `VIDEO\_LOCALITY\_MODEL.md`.

The hard part (the doc's centerpiece)

The **de-WSL native-compute port**: today WASB/TrackNet run in **WSL2** on an NVIDIA GPU (CUDA) ? Windows-specific. The port must run the detector **natively per-OS**: Linux+NVIDIA ? CUDA directly; macOS (Apple Silicon, no CUDA) ? MPS/CPU; Windows native (no WSL). It's grounded in the existing `DetectorRunner` ABC, which already documents a Linux-native runner as "the main de-risk goal."

The make-or-break unknown is compute on a **typical no-GPU enthusiast laptop**. WASB is ~3.4x real-time on a laptop RTX 2070 Super (measured); on CPU it's far slower, and Apple has no CUDA. The doc names this the gating question and proposes a **P0 feasibility spike** to measure CPU/MPS throughput **before any app is built**.

Structure (per `PROJECT\_DOC\_TEMPLATE.md`)

Status header · decisions-locked (D1?D6) · foundation/research · architecture + reuse map · risk table · honest limits · ****\$7 progress tracker as source-of-truth**** · open questions · definition-of-done.

****\$7 tracker:**** `P0 compute-feasibility spike ? P1 de-WSL native detector ? P2 desktop app shell (Tauri vs Electron vs CLI) ? P3 packaging/code-signing per OS`. P0 is the only un-gated row; P1?P3 are owner-greenlit *after* P0's verdict.

Honest limits

Feasibility hinges entirely on the compute question. If CPU-only is too slow on commodity laptops, the simple "wrap the pipeline" path doesn't exist ? it becomes "ship a smaller exported owned model" (ties `OWNED\_MODEL\_IMPLEMENTATION\_PLAN`) or a GPU-tier-only product. The doc says this plainly.

Scope of this PR

Docs only (`docs/DESKTOP\_APP\_PLAN.md` + `docs/README.md`). ****No implementation.**** Per CLAUDE.md, the desktop app is owner-gated; the scoping doc + (if greenlit) the P0 spike are the deliverable.

? Generated with [Claude Code](https://claude.com/claude-code)

50. user (2026-06-18T07:49:51.433Z)

<https://github.com/avidullu/badminton-highlight-indexer/pull/219>

51. assistant (2026-06-18T07:50:05.891Z)

PR is open: [#219](https://github.com/avidullu/badminton-highlight-indexer/pull/219). Let me record a micro-checkpoint in the live memory (per the checkpoints working agreement) ? a new tracked subproject + PR is a meaningful checkpoint.

52. user (2026-06-18T07:50:22.344Z)

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>

53. assistant (2026-06-18T07:50:27.189Z)

The memory file changed under me (concurrent session). Let me re-read the top and re-apply.

54. user (2026-06-18T07:50:27.765Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-18 (? **R1 MILESTONE LAUNCHED ? #204 MERGED** `67b6927`; flip
re-verified at **n=13**, owner-approved). Owner: "verify #204 still stands at n=13, merge if
findings stand" ? it stands (stronger), verified 3 ways, merged. **Local-windowing defaults now
ON in production: cap=6 + R4** (`max_window_duration 6`, `window_inactivity_end 1.5`,
`inactivity_rally_thresh 0.20`, `inactivity_min_density 0.30`). W2 / W1.5 hard-cap / R5 stay
```

default-OFF.

```
13      - ? **n=13 re-verify (2 new Boxhill clips GX010137/GX010141):** baseline?milestone tolF1
**?+0.222, 12/0/1, p=0.0005** (stronger than n=11's 10/0); cap6 beats cap12 **12/0/1,
p=0.0005**; cap6?+R4 +0.040, **9/1/3**, p=0.0215 (R4's 1 loss = GX010137, a tolerance artifact ?
strict f1 & |?end| both IMPROVE there). Net guard **PASS**, merge_rate 0.651?0.161, **0 per-
video regressions**. Both new clips improve (0.448?0.685, 0.333?0.431). tIoU ladder 0.236?0.474.
14      - ? **Corroborated 2 ways:** sibling **#214 `lovo_resweep.py`** matches to 3dp + LOVO
cap=6 **13/13** on tIoU; my **5-lens adversarial workflow** `wf_66c34d9d-057` ? **STANDS**
(every L00 fold clears; worst-fold drop GX010128 still ?+0.202 p=0.0010 ? no single clip drives
it). Reproduce: `scratch/r1_evidence.py`.
```

55. user (2026-06-18T07:50:43.970Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

56. assistant (2026-06-18T07:50:49.282Z)

Let me do a final verification ? PR state and that the working tree is clean apart from the pre-existing sibling WIP.

57. user (2026-06-18T07:50:52.336Z)

```
---PR---
{"base":"master","files":["docs/DESKTOP_APP_PLAN.md","docs/README.md"],"head":"docs/desktop-app-plan","number":219,"state":"OPEN","title":"docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port"}
---WORKING TREE (sibling WIP + untracked only; scratch msgs gone)---
M backend/eval/rally_seg_eval.py
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

58. assistant (2026-06-18T07:51:11.968Z)

Done. The tracked subproject is scoped and the PR is open.

What landed ? [PR #219](https://github.com/avidullu/badminton-highlight-indexer/pull/219) (docs-only, owner)

****New doc:**** docs/DESKTOP_APP_PLAN.md, indexed one-line in docs/README.md under a new "Product / platform plans" section. Follows `PROJECT\_DOC\_TEMPLATE` (status header · D1?D6 decisions-locked · \$7 tracker as source-of-truth · definition-of-done).

****The vision it scopes:**** package the mature local pipeline (WASB + improved windowing + ffmpeg stitch, already runnable via `generate\_highlights.py --skip-ai-handoff`) as a cross-platform desktop app for non-technical players ? *plug in the card, click once, get back only the reels*, fully on-device, no upload, no original-video bloat. It's framed as realizing the ****L0 fully-local**** tier of [VIDEO_LOCALITY_MODEL.md](docs/VIDEO_LOCALITY_MODEL.md) and the ****LocalDesktop ComputeBackend**** of [PLATFORM_ARCHITECTURE.md](docs/PLATFORM_ARCHITECTURE.md) §3.2.

****The centerpiece (the real "port to Linux/Mac" subproject):**** the de-WSL native-compute port. I grounded this in code ? backend/pipeline/detectors/base.py already documents a `LinuxNativeRunner`/`ONNXRunner` as ****the main de-risk goal,**** the WSL plumbing is isolated in a separate mixin, and the torch inference core in [wasb_infer.py](backend/pipeline/detectors/wasb_infer.py) just needs lifting out of WSL (identical CSV contract ? windowing/stitch untouched).

****The make-or-break unknown, made the doc's spine:**** compute on a typical ***no-GPU*** enthusiast laptop. WASB is ~3.4× real-time on a laptop 2070S (measured); CPU is far slower, Apple has no CUDA. ****P0** is a feasibility spike that measures CPU/MPS for a ~10-min clip and returns go/no-go before any app code ? and explicitly names the fork: if CPU-only is too slow, the path needs a smaller exported owned model (ties [OWNED_MODEL_IMPLEMENTATION_PLAN.md](docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md)) or a tiered

fast/slow design.

****§7 tracker:**** P0 compute spike ? P1 de-WSL native detector ? P2 app shell (Tauri vs Electron vs CLI+installer) ? P3 packaging/code-signing per OS. Only P0 is un-gated; P1?P3 are owner-greenlit *after* P0. Licensing locked to MIT/Apache/BSD + an LGPL ffmpeg build (no GPL x264).

Process notes

- Branched off `origin/master` (`a010667`) per the branching-safety convention; staged ****explicit paths only****.
- ?? A sibling/owner WIP modification to `backend/eval/rally\_seg\_eval.py` appeared in the shared checkout mid-task ? I left it ****untouched**** (not staged, not committed).
- (Minor: my first commit's message got mangled because I used PowerShell here-string syntax inside the Bash tool; amended cleanly via `-F`.)
- Recorded a micro-checkpoint in `memory/current-state.md`.

****Owner gate stands:**** this is the scoping doc only. The P0 spike ? and anything past it ? needs your explicit greenlight before I build.